Supplementary Material

PKPD Model Equations

Pharmacokinetics

To describe the pharmacokinetics of a hemoglobin binder (HbS), including PF-07059013, a pharmacokinetic model was built and consisted of four compartments: oral (dosing compartment), plasma, RBC, and peripheral compartments.

To simulate oral dosing, drug absorption from the oral (depot) to plasma compartments is modeled with an absorption rate constant k_a. The differential equation for the amount of drug in the oral compartment, A_o, is,

$$\frac{dA_o}{dt} = -k_a . A_{o.}$$

Drug in the plasma compartment can be exchanged between plasma and RBC and peripheral compartments and is eliminated at a rate (CL_p).

$$\frac{dA_p}{dt} = k_a . A_o - CL_p C_p - Q_{per}(C_p - C_{per}) - Q_{rbc}(C_{p,free} - C_{rbc}),$$

where A_p is amount in the plasma compartment, and $C_p = A_p/V_p$, is the concentration; the volume of the plasma compartment is defined as $V_p = \left(1 - \frac{HCRIT}{100}\right) V_{blood}$, where HCRIT corresponds to hematocrit (assumed to be constant in healthy participants) and V_{blood} is the total blood volume. Q_{per} and Q_{rbc} are the flow rates between the peripheral and red blood cell compartment respectively. A scaled plasma concentration, with a constant fraction unbound so that $C_{p,free} = fu. C_p$, is assumed for exchange between the plasma and RBC compartment.

The differential equations for drug amount in the peripheral and RBC compartments are given by,

$$\frac{dA_{per}}{dt} = Q_{per}(C_p - C_{per})$$

$$\frac{dA_{rbc}}{dt} = Q_{rbc}(C_{p,free} - C_{rbc}) - V_{rbc}(2\, k_{on1}^{Hb}.C_{rbc}.C_{Hb0} - k_{off1}^{Hb}C_{Hb1}) -$$

$$V_{rbc}(k_{on2}^{Hb}.C_{rbc}.C_{Hb1} - 2\, k_{off2}^{Hb}C_{Hb2}) - \mathrm{k_{death}}\left(\frac{\mathrm{N_{rbc}}}{\mathrm{N_{ret+N_rbc}}}\right)\mathrm{A_{rbc}}.$$

Hb binding to PF-07059013 in the RBC compartment was described by two successive binding reactions (with dissociation constants $K_{on1}^{Hb} = \frac{k_{off1}^{Hb}}{k_{on1}^{Hb}}$ and $K_{on2}^{Hb} = \frac{k_{off2}^{Hb}}{k_{on2}^{Hb}}$). To simulate the pharmacology in healthy participants, an approximation of $\frac{\mathrm{N_{rbc}}}{\mathrm{N_{ret+N_rbc}}} \approx 1$ was made as a typical percent reticulocyte in healthy participants is less than 2%. Finally, the volume of the red blood cell compartment is defined as $V_{rbc} = \frac{HCRIT}{100}.V_{blood}$. To describe PK in healthy participants, RBC turnover is not considered ($k_{death} = 0$).

Amounts of free and drug-bound hemoglobin are tracked using the following equations,

$$\frac{dA_{Hb0}}{dt} = J_{Hb}, synthesis - V_{rbc}(2\, k_{on1}^{Hb}.C_{rbc}.C_{Hb0} - k_{off1}^{Hb}C_{Hb1}) - k_{death}\left(\frac{N_{rbc}}{N_{ret+N_rbc}}\right)A_{Hb0}$$

$$\frac{dA_{Hb1}}{dt} = V_{rbc}(2\, k_{on1}^{Hb}.C_{rbc}.C_{Hb0} - k_{off1}^{Hb}C_{Hb1}) - V_{rbc}(k_{on2}^{Hb}.C_{rbc}.C_{Hb1} - 2\, k_{off2}^{Hb}C_{Hb2})$$

$$- k_{death}\left(\frac{N_{rbc}}{N_{ret+N_{rbc}}}\right)A_{Hb1}$$

$$\frac{dA_{Hb2}}{dt} = V_{rbc}(k_{on2}^{Hb}.C_{rbc}.C_{Hb1} - 2\, k_{off2}^{Hb}C_{Hb2}) - k_{death}\left(\frac{N_{rbc}}{N_{ret+N_{rbc}}}\right)A_{Hb2}$$

$J_{Hb}, synthesis$ corresponds to the synthesis rate of Hemoglobin. For healthy participants, the total Hemoglobin is assumed to be conserved overtime, hence $J_{Hb}, synthesis = k_{death}.V_{rbc}.Hb_{rbc}^{tot}$ where Hb_{rbc}^{tot} is the Hb concentration in a red blood cell (Mean corpuscular hemoglobin concentration); however, as RBC turnover is not considered for healthy participants ($k_{death} = 0$ and $J_{Hb}, synthesis = 0$).

The dissociation rate constant for drug binding to unbound Hb (Kd_1) was fixed and the Kd for drug binding to drug-bound Hb (Kd_2) was estimated to be much smaller in value, indicating cooperative binding of the drug to drug-bound Hb. Hb coverage was calculated as the ratio of drug bound to Hb to the total Hb (i.e., sum of drug bound and unbound Hb in RBCs),

$$Hb_{cov} = \left(\frac{A_{Hb1} + A_{Hb1}}{A_{Hb0} + A_{Hb1} + A_{Hb2}}\right).100\%$$

See Table S2 for a summary of the parameter estimates for the PK model.

Semi-mechanistic RBC model

A semi-mechanistic model of erythropoiesis was built with the following assumed regulatory processes:

1. Regulation of RBC production – the rate of erythroid progenitor differentiation increases with serum erythropoietin (EPO) level.

2. Regulation of EPO – EPO production rate decreases with total circulating RBC population.

3. Regulation of RBC elimination – RBC hemolysis rate decreases as the fraction of HbS bound to drug increases.

EPO dynamic is described by the following equation,

$$\frac{dEPO}{dt} = V_{EPO} - k_{el}^{EPO}.EPO$$

where EPO production is dependent on the total number of RBCs,

$$V_{EPO} = V_{EPO}^{max}\left(\frac{K_{RBC}^{n}}{K_{RBC}^{n} + (N_{RBC} + N_{RET})^{n}}\right).$$

In turn, EPO dynamics determine the birth rate of colony forming units N_{CFUe} as described by the following equations,

$$\frac{dN_{CFUe}}{dt} = k_{birth}^{CFUe} - k_{CFUe,Ret}\,.N_{CFUe},$$

$$k_{birth}^{CFUe} = V_{R}^{min}\left(1 + \frac{Emax_{EPO}\,EPO^{m}}{K_{EPO}^{m} + EPO^{m}}\right).$$

As $k_{CFUe,Ret}\,.N_{CFUe}$ describes the rate of reticulocyte production, a proportional term is used to the determine the rate of hemoglobin synthesis, i.e. for sickle cell disease population,

$$J_{Hb}, synthesis = k_{CFUe,Ret}\,.N_{CFUe}.V_{rbc}.Hb_{rbc}^{tot}.$$

Finally, we utilize a two transit-compartment model to describe the maturation of reticulocyte into red blood cells as described using the following equations,

$$\frac{dN_{RET}}{dt} = k_{CFUe,Ret}\,.N_{CFUe} - k_{ret}N_{ret},$$

$$\frac{dN_{RBC}}{dt} = k_{ret}\,N_{ret} - k_{death}N_{RBC}.$$

Through its mechanism of action, PF-07059013 is assumed to improve red blood cell lifetime by preventing sickling. As a simplifying assumption, we take k_{death} to instantaneously change with Hb coverage,

$$k_{death} = k_{death0}\left(1 - \frac{I_{max}\,{Hb_{cov}}^{nd}}{{K_{cov}}^{nd} + {Hb_{cov}}^{nd}}\right),$$

where k_{death0} is the baseline red blood cell death rate.

PKPD Simulation Assumptions

The key difference in the two sets of populations, i.e., healthy and SCD individuals was the range of hematocrit for these populations. The hematocrit levels for healthy population were set to be a mean of 43% (with a range of 36 – 50%), whereas they were set to a mean of 27% (with a range of 15 – 35%) for the SCD population.

In the linear mixed-effects PK/PD model linking population PK predicted Hb coverage with change in p20 assumed a zero intercept, i.e., no effect on p20 without any Hb coverage. For calculating the 90% CI of change in p20, Hb coverage was based on a dose of 2400 mg BID, which had the closest AUC/Cmax ratio compared to 4000 mg total daily dose.

Virtual Population Simulation for Semi-Mechanistic RBC

To characterize heterogeneity in the SCD patient population, we performed a virtual population simulation[1] that incorporated pooled subject-level data from MSH[2-3] to reflect hematology variability (specifically, RBC count, % Reticulocyte, Hb, mean corpuscular volume, and % fetal hemoglobin) in the population at baseline. Additionally, EPO range in SCD subjects is based on values listed in Ballas and Marcolina[4]. Parameter values for each virtual population were obtained using steady state constraint, or estimated to be within physiologically relevant ranges. Baseline RBC lifetime was adjusted for fetal hemoglobin (HbF) level[5] for each individual virtual patient (HbF ranges obtained from MSH study); a linear relationship between RBC and %HbF was assumed by fitting to data shown in Quinn et al.[5] Drug effect was incorporated into the model by assuming that RBC lifetime increased with Hb coverage as described by a sigmoidal Emax model as shown in Figure 4(A). Due to lack of clinical data to constrain model parameters,

we assumed a large uncertainty in parameters describing RBC lifetime so that maximum RBC lifetime (at high Hb coverage) ranges from 50-100 days and EC50 ranges from 5-30% Hb coverage. Parameter ranges used in the virtual population simulation are listed in Table S3.

Data

The FIH healthy participant study (NCT04323124) was conducted in accordance with the International Council for Harmonisation Guideline for Good Clinical Practice and the Declaration of Helsinki. All participants were required to provide written informed consent before any study-related procedures and the protocol, amendments, and informed consent forms were approved by the institutional review board or independent ethics committee.

Supplementary figures and tables

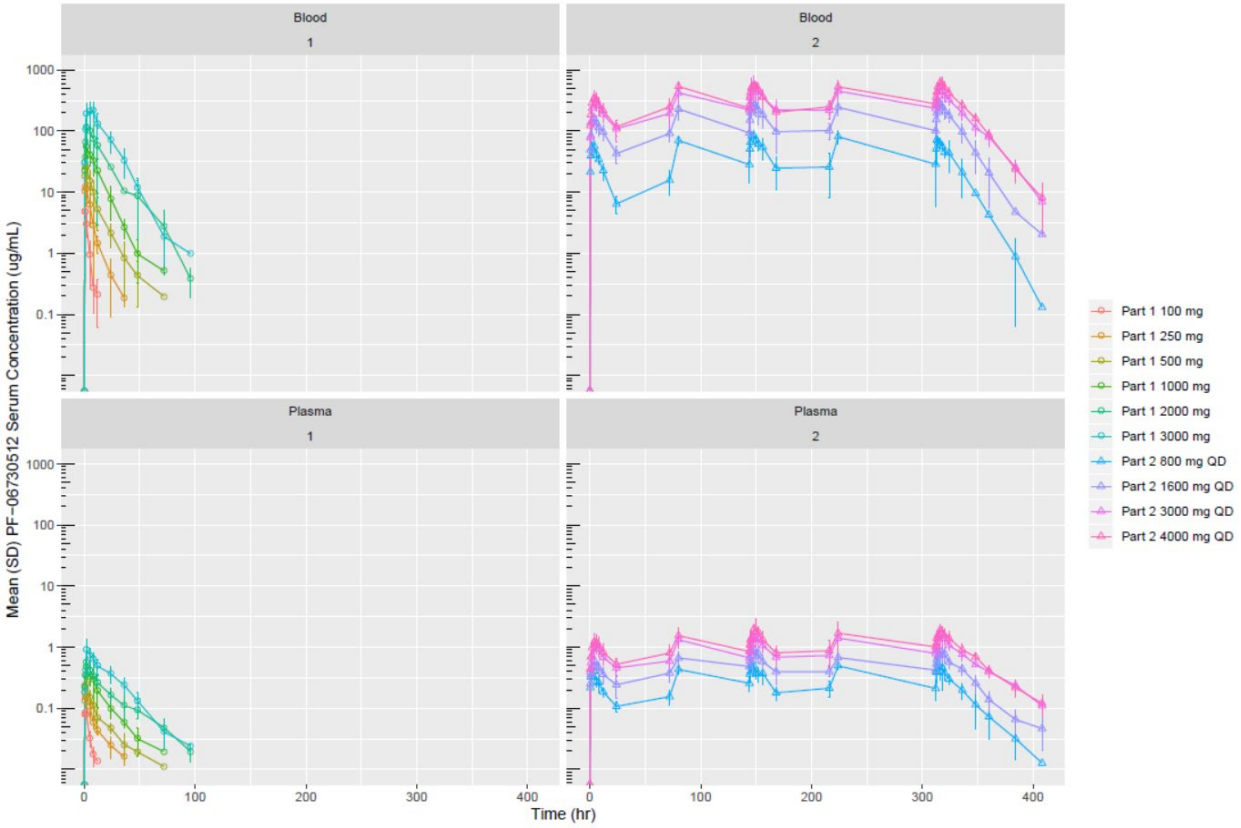

Figure S1: Mean (+ SD) blood and plasma PF-06730512 concentration-time profiles following single and multiple doses . Part 1 represents the single ascending dose part of the study and Part 2 represents the multiple ascending dose part of the study.

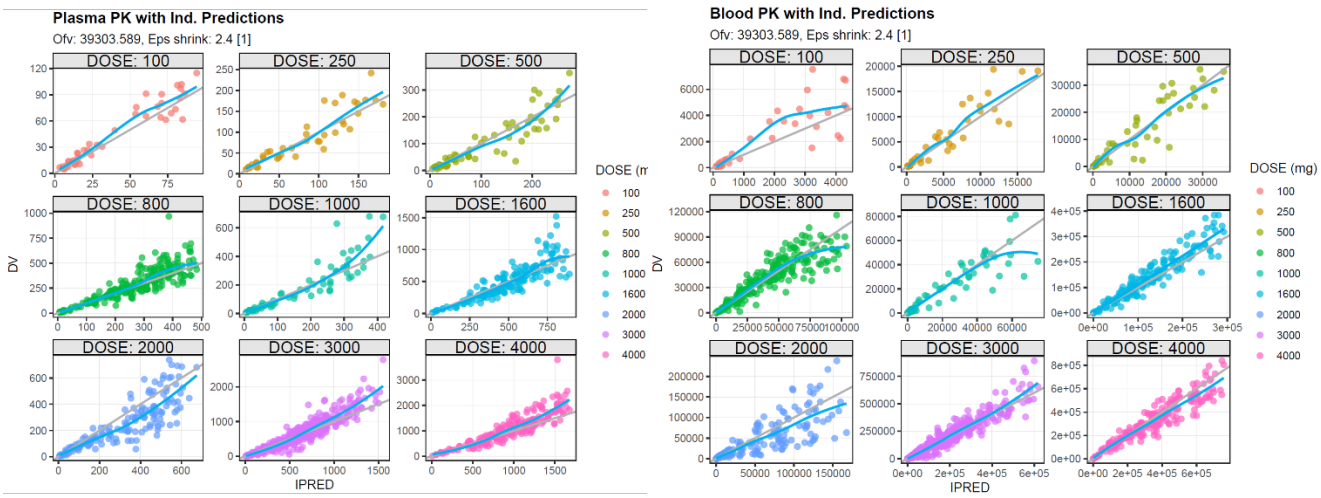

Figure S2: Goodness of Fit plots for Drug Concentration (PK) in Plasma (left) and in Blood (right). Drug Concentration is shown on the y-axis as DV (Dependent Variable) and is faceted by different doses (in mg).

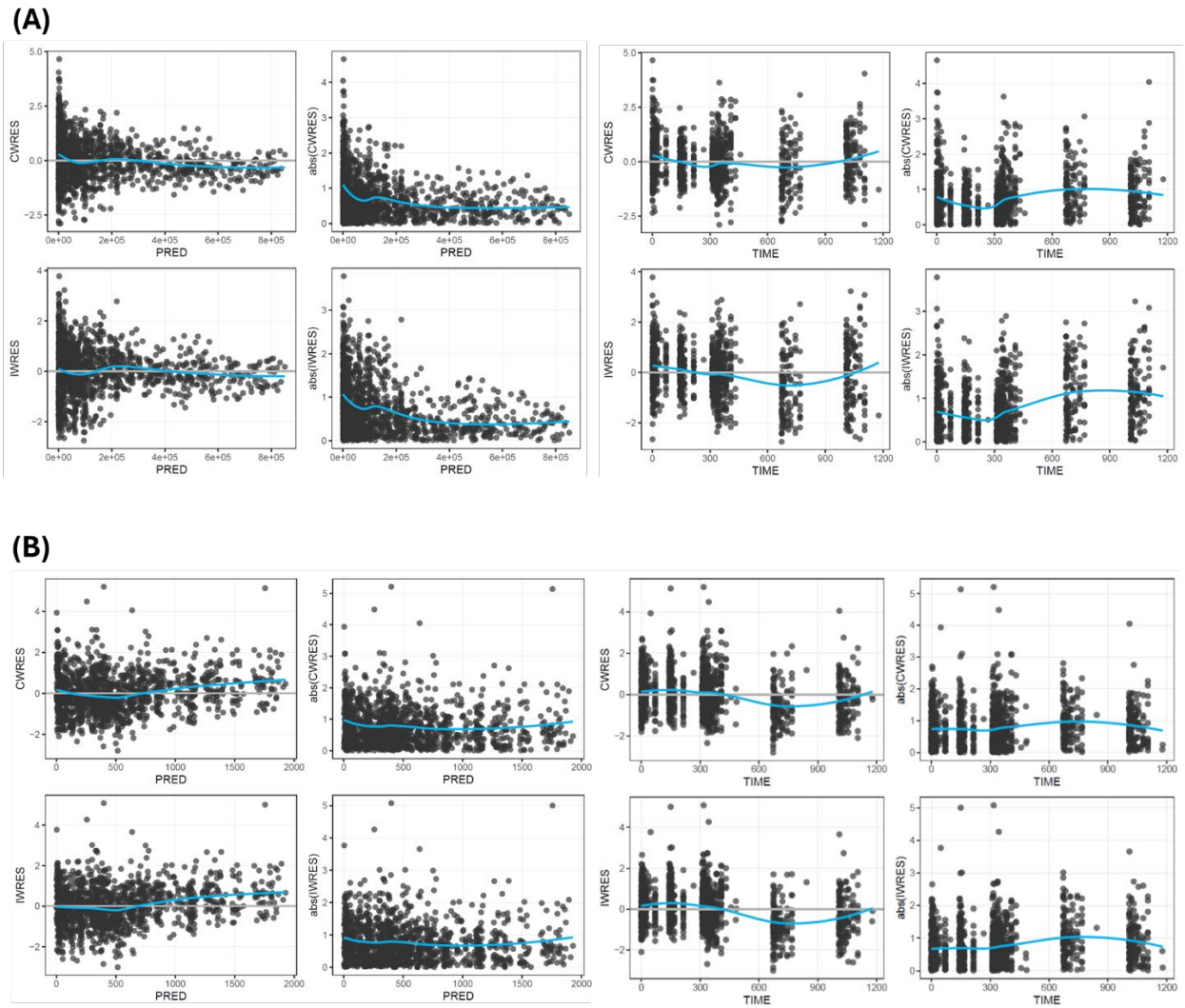

Figure S3: Panel (A) - Residuals with Population Predictions (left) and with independent variable, time (right) for Concentrations in Blood. Panel (B) - Residuals with Population Predictions (left) and with independent variable, time (right) for Concentrations in Plasma. Conditional Weighted Residuals (CWRES), Individual Weighted Residuals (IWRES) and their absolute values abs(CWRES), abs(IWRES) are shown with population predictions (left panel) and with time (right panel).

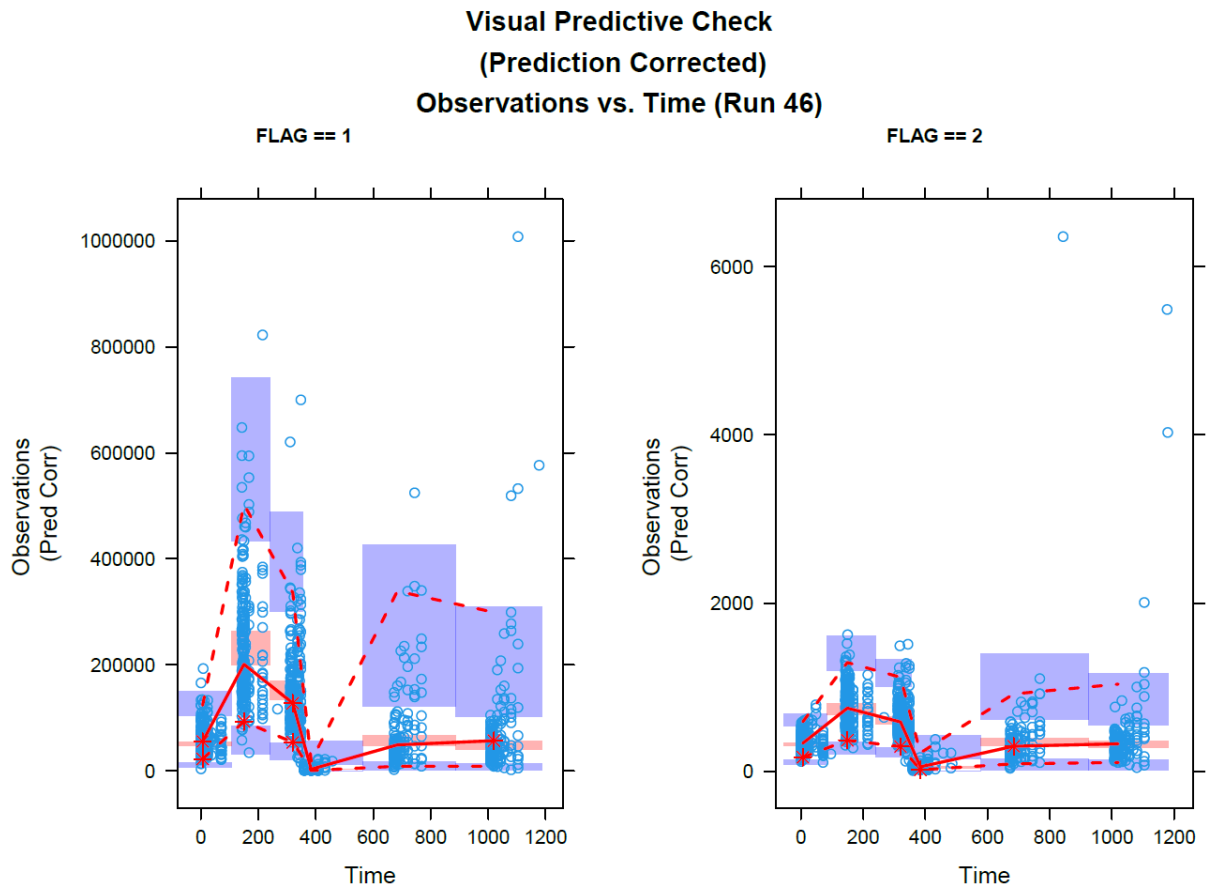

Figure S4: Prediction Corrected VPC (Visual Predictive Check) for Plasma and Blood Concentrations. Left Panel (FLAG == 1) shows the prediction corrected VPC for drug concentration in Blood. Right Panel (FLAG == 2) shows the same for the drug concentration in Plasma. N = 2000 samples were simulated using PsN VPC (version 5.5.0) to generate the figure. The solid red line denotes median of observed values (prediction corrected) whereas the dashed red lines show the 5th and 95th percentile of the same. The red bands show 95% confidence interval of the median values from simulated data (prediction corrected) and the purple bands show the

95% confidence interval around the 5th and 95th percentiles from the simulated data (prediction corrected). The blue circles show prediction corrected observed values.

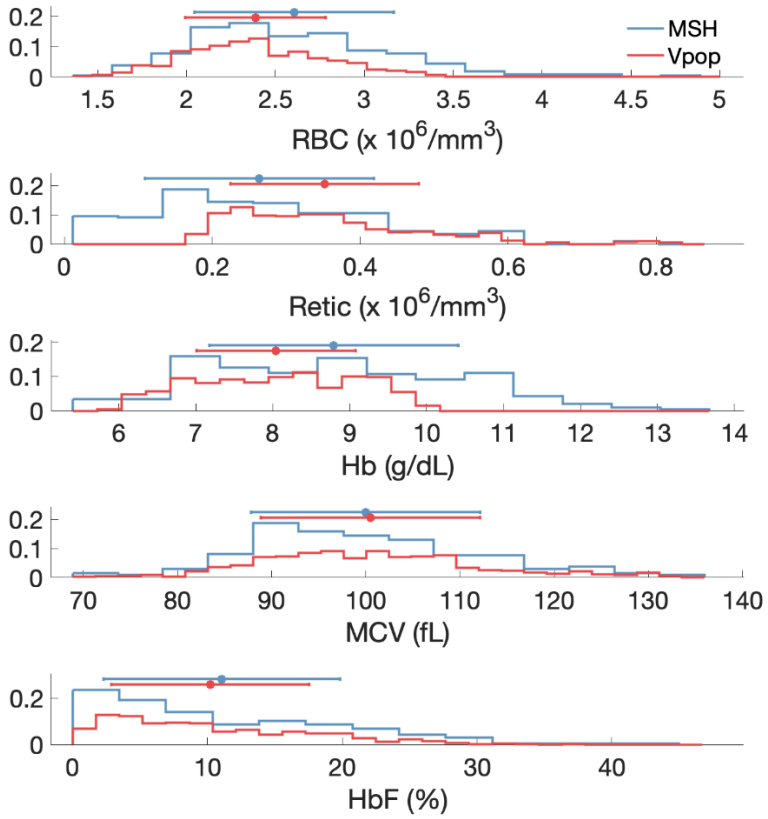

Figure S5: Comparison of Baseline Distribution between Virtual Population (blue, N=421) and MSH adult sickle-cell subject data[2] (red, N=408). A planned inclusion-exclusion criteria for the planned SCD patient study was applied for the virtual population simulation (Hb between 6-10 g/dL and Reticulocyte Count greater than 1.5 times upper limit of normal at screening).

Table S1: Demographic Table by Study Cohorts

Baseline Characteristic	Statistic	Part 1[a]		Part 2[b]				Combined
		Cohort 1	Cohort 2	Cohort 3	Cohort 4	Cohort 5	Cohort 6	
Age (years)	N	9	12	10	9	8	8	56
	Mean	41.7	38.8	30.5	30.6	27.6	33.4	34.1
	SD	6.8	10.1	8.4	10.6	8.5	9.5	10
	Median	40	38	26.5	28	27	32.5	31
	Min	34	24	21	21	20	23	20
	Max	52	53	45	51	45	45	53
Weight (kg)	N	9	12	10	9	8	8	56
	Mean	77.7	77.1	77.2	75.9	75.6	74.7	76.5
	SD	8.3	9.2	13.3	9.5	13.7	10.6	10.4
	Median	75	78.5	76.8	76.1	72.9	76.1	76.2
	Min	66	54.8	56.1	61	61.5	55	54.8
	Max	92.2	92.2	101	91.1	95	86.1	101
Hematocrit (%)	N	9	12	10	9	8	8	56
	Mean	42.3	43.1	43.7	44	43.2	43	43.2
	SD	2.9	1.9	2	2.9	2.8	3.1	2.5
	Median	42	42	44	44	42	43.5	43
	Min	37	41	41	39	41	38	37
	Max	47	47	47	48	49	47	49
Sex	Total	9	12	10	9	8	8	56
	Male	9	12	10	9	8	8	56
	Female	0	0	0	0	0	0	0

[a] Participants from Cohorts 1 and 2, in Study Part 1, received single ascending dose of 100 mg to 3000 mg, or placebo.

[b] Participants from Cohorts 3 to 6, in Study Part 2, received multiple ascending doses of 800 mg, 1600 mg, 3000 mg, and 4000 mg, or placebo QD for 14 days respectively.

Table S2A: Final Parameter Estimates of the Population PK Model (THETAs) to Characterize Plasma and Blood Drug Concentrations

Parameter Description (units) – Population Parameters (Theta)	**Final Estimate**	**Bootstrap Median (95% CI)**	**Fixed**
Clearance from Plasma Compartment (L/hr)	85	84.96 (68.65, 117)	No
Intercompartmental Flow Rate from Plasma to Peripheral Compartment (L/hr)	132	130.78 (71.6, 248.18)	No
Volume of Peripheral Compartment (L)	1200	1202.2 (936.41, 1849.76)	No
Volume of Blood Compartment (L)	5	5	Yes
Bioavailability (Unitless, between 0 and 1)	0.608	0.61 (0.45, 0.87)	No
Rate of absorption from the Depot to Plasma Compartment (1/hr)	0.666	0.66 (0.46, 1.04)	No
Proportional Error on Plasma Concentration (Unitless)	0.296	0.29 (0.27, 0.32)	No
Additive Error on Plasma Concentration (μM)	1E-03	1E-03	Yes
Proportional Error on Blood Concentration (Unitless)	0.35	0.35 (0.3, 0.39)	No
Additive Error on Blood Concentration (μM)	1E-03	1E-03	Yes
Kd for the first binding reaction (μM)	68.6	68.6	Yes
Kd for the second binding reaction (μM)	9.92E-03	0.01 (0.01, 0.01)	No
Proportional Decrease in bioavailability for formulation without polymer	1.02E-02	0.01 (-0.12, 0.15)	No

Proportional Decrease in absorption rate for formulation without polymer	0.416	0.41 (0.32, 0.53)	No
Proportional Increase in bioavailability due to food effect	-0.263	-0.25 (-0.59, -0.06)	No
Proportional Decrease in absorption rate due to food effect	0.343	0.34 (0.15, 0.54)	No
Fraction unbound for plasma concentration (Unitless)	0.162	0.162	Yes
Flow Rate to RBC Compartment (L/hr) – assumed rapid distribution	1E5	1E5	Yes
Binding on rate for drug to Hb (kon for both sites) (1/μM.hr)	7200	7200	Yes

Bootstrap parameter estimates were calculated using N = 1000 samples from a PsN Bootstrap routine of the final model.

Table S2B: Final Parameter Estimates of the Population PK Model (OMEGAs) to Characterize Plasma and Blood Drug Concentrations

Parameter Description (units) – InterIndividual Variability (IIV) Parameters (Omega)	**Final Estimate (Variance)**	**Bootstrap Median (95% CI)**	**Shrinkage (Std. Dev.)**
IIV on clearance from Plasma compartment	4.52E-02	0.04 (0.03, 0.08)	0.629%
IIV on Intercompartmental Flow Rate from Plasma to Peripheral Compartment	0.364	0.33 (0.06, 1.36)	13.542%
IIV on volume of Peripheral Compartment	0.167	0.16 (0.09, 0.25)	3.982%

Rate of absorption from the Depot to Plasma Compartment	9.21E-02	0.09 (0.05, 0.14)	9.515%

Bootstrap parameter estimates were calculated and reported above using N = 1000 samples from a PsN Bootstrap routine of the final model. Sampling Importance Resampling (SIR) was also performed, and although SIR produced slightly narrower 95% confidence intervals, the values from the bootstrap were retained as the more conservative approach.

Table S3: Semi-mechanistic RBC model parameter definitions and ranges used in virtual population simulation

Parameter	Definition	Unit	Median	90% Range	Source
k_{death0}	Baseline RBC death rate	1/hr	0.0015	0.0011–0.0019	Based on %HbF [2][a]
I_{max}	Imax for RBC death rate		0.78	0.50 – 0.87	Assumed max RBC lifetime: 50-200 days
K_{cov}	EC50 for RBC death rate function	%	25.8	11.3 – 38.8	Based on initial fitting to voxelotor data[d]
nd	Hill coefficient for RBC death rate function		2	Fixed	
V_{EPO}^{max}	Maximum EPO production flux	U/mL.hr	0.38	0.026 – 20.2	Estimated to match EPO range [4][c]
K_{RBC}	EC50 for EPO production function	10^{12} cells	5.11	2.33 – 7.70	Assumed[e]
n	Hill coefficient for EPO production function		5	Fixed	
k_{el}^{EPO}	EPO elimination rate	1/hr	0.038	0.024 – 0.079	EPO half-life 8-24 hour [7]
V_{rbc}	RBC volume (Mean corpuscular volume)	L/cell	9.96e-14	8.38e-14 – 1.23e-13	Based on MCV [2]
Hb_{rbc}^{tot}	Hb concentration in an RBC (Mean corpuscular hemoglobin concentration)	uM	4630	4070 – 5148	Based on Hb [2][b]
V_R^{min}	Minimum CFUe production	10^{12} cells/hr	0.0015	0.00019-0.011	Steady state constraint
$Emax_{EPO}$	Emax for CFUe production function		129.8	18.7-234	Estimated[f]
K_{EPO}	EC50 for CFUe production function	U/mL	0.32	0.070-0.53	Estimated[f]

m	Hill coefficient for CFUe production function		1.5	Fixed	
$k_{CFUe,Ret}$	CFUe to Reticulocyte transition rate	1/hr	0.0042	0.0022 – 0.022	Assumed between 0.5-14 days [6]
k_{ret}	Reticulocyte to RBC transition rate	1/hr	0.011	0.0051-0.020	Based on Reticulocytes count [2][b]

a) Assumed relationship RBC Lifetime (days) = 21.6 + 0.64*%HbF based on regression fitting of data [5], kdeath0 = log(2)/RBC Lifetime.
b) Steady-state constraint for each virtual patient, to match RBC, Hb, Retic distribution seen in MSH data [2].
c) Steady-state constraint based on virtual patient's EPO value. Assumed normal EPO distribution based on data in [4], 74.4±36.9 mU/mL.
d) Estimated based on parameter fitting to Phase 3 Voxelotor data [8] resulting in a range of ~15-35%. Estimated threshold for %HbF protection is ~25% [9].
e) Values chosen to vary - observed range of total RBC count in SCD based on MSH data is 6-20x10^{12} cells.
f) Initial estimation based on partial calibration to phlebotomy data in healthy adults [10]. Virtual population parameters chosen over uniform distribution over one order of magnitude to reflect uncertainty in parameter.

References

1. Allen RJ, Rieger TR, Musante CJ. Efficient Generation and Selection of Virtual Populations in Quantitative Systems Pharmacology Models. *CPT Pharmacometrics Syst Pharmacol*. 2016;5(3):140-146. doi:10.1002/psp4.12063

2. Multicenter Study of Hydroxyurea (MSH) [Dataset] https://biolincc.nhlbi.nih.gov/studies/MSH/

3. Charache S, Terrin ML, Moore RD, et al. Effect of hydroxyurea on the frequency of painful crises in sickle cell anemia. Investigators of the Multicenter Study of Hydroxyurea in Sickle Cell Anemia. N Engl J Med. 1995;332(20):1317–22. doi: 10.1056/NEJM199505183322001.

4. Ballas SK & Marcolina MJ. Determinants of Red Cell Survival and Erythropoietic Activity in Patients with Sickle Cell Anemia in the Steady State. Hemoglobin 2000, 24(4), 277–286.

5. Quinn CT, Smith EP, Arbabi S, et al. Biochemical surrogate markers of hemolysis do not correlate with directly measured erythrocyte survival in sickle cell anemia. Am J Hematol. 2016;91(12):1195-1201.

6. Shilpa M. Hattangadi, Piu Wong, Lingbo Zhang, Johan Flygare, Harvey F. Lodish; From stem cell to red cell: regulation of erythropoiesis at multiple levels by multiple proteins, RNAs, and chromatin modifications. *Blood* 2011; 118 (24): 6258–6268

7. Halstenson, C.E., Macres, M., Katz, S.A., Schnieders, J.R., Watanabe, M., Sobota, J.T. and Abraham, P.A. (1991), Comparative pharmacokinetics and pharmacodynamics of epoetin alfa and epoetin beta. Clinical Pharmacology & Therapeutics, 50: 702-712.

8. Vichinsky E, Hoppe CC, Ataga, KI, Ware RE, Nduba V, El-Beshlawy A, Hassab H, Achebe MM, Alkindi S, Brown RC and Diuguid DL. A phase 3 randomized trial of voxelotor in sickle cell disease. N Engl J Med. 2019; *381*(6), 509-519.

9. Powars DR, Weiss JN, Chan LS, Schroeder WA. Is there a threshold level of fetal hemoglobin that ameliorates morbidity in sickle cell anemia? Blood 1984; 63, 921-926.

10. Gleiter CH, Freudenthaler S, Delabar U, Eckardt KU, Muhlbauer B, Gundert-Remy U, and Osswald H. Erythropoietin production in healthy volunteers subjected to controlled haemorrhage: evidence against a major role for adenosine. Brit J Clin Pharmacol, 1996; 42: 729-735.

Code for Figure 3 A and B

```
### PK simulation for HbS
library(ggplot2)
library(tidyverse)
library(dplyr)
library(scales)
library(RColorBrewer)
library(RxODE)
library(mvnfast)
library(pracma)
library(gridExtra)
library(reshape)
library(PKNCA)

theme_bwMod1 <- function() {
  a <- theme(panel.background = element_rect(fill = "white", colour = NA),
            panel.border =      element_rect(fill = NA, colour="black", size = 0.2),
            panel.grid.major =  element_line(colour = "grey90", size = 0.2),
            panel.grid.minor =  element_line(colour = "grey98", size = 0.5),
            strip.background =  element_rect(fill = "grey80", colour = "grey50", size = 0.1),
            strip.text.x =      element_text(size = 10 * 0.8),
            strip.text.y =      element_text(size = 10 * 0.8, angle = -90),
            axis.text.x =
element_text(colour="black",size=10,angle=0,hjust=.5,vjust=.5,face="plain"),
            axis.text.y =
element_text(colour="black",size=10,angle=0,hjust=1,vjust=0.5,face="plain"),
            axis.ticks = element_line(colour = "black", size = 0.2),
            plot.title = element_text(size = 10),
            plot.subtitle = element_text(size = 8),
            plot.caption = element_text(size = 8, face = "italic"))
}


theme_bwMod2 <- function() {
  a <- theme(panel.background = element_rect(fill = "white", colour = NA),
            panel.border =      element_rect(fill = NA, colour="black", size = 0.2),
            panel.grid.major =  element_line(colour = "grey90", size = 0.2),
            panel.grid.minor =  element_line(colour = "grey98", size = 0.5),
            strip.background =  element_rect(fill = "grey80", colour = "grey50", size = 0.1),
            strip.text.x =      element_text(size = 10 * 0.8),
            strip.text.y =      element_text(size = 10 * 0.8, angle = -90),
            axis.text.x =
element_text(colour="black",size=10,angle=0,hjust=.5,vjust=.5,face="plain"),
```

```
            axis.text.y =
element_text(colour="black",size=10,angle=0,hjust=1,vjust=0.5,face="plain"),
            axis.ticks = element_line(colour = "black", size = 0.2),
            plot.title = element_text(size = 10),
            plot.subtitle = element_text(size = 8),
            plot.caption = element_text(size = 8, face = "italic"),
            legend.title = element_blank())
}

# ggplot color scale
gg_color_hue <- function(n) {
  hues = seq(15, 375, length = n + 1)
  hcl(h = hues, l = 65, c = 100)[1:n]
}

runno = list.files(pattern=".ext")
mod =  strsplit(runno,'[.]')[[1]][1]
modelCode <- "HbS_2cptMechPKModel1"
projRun <- paste0(modelCode,"_",mod)

simHbSPKmodel1 <- RxODE({

  Dose_mg_ = DoseVal_mg_;

  # Assign Individual PK parameters - based on HV SAD PK parameter estimates
  CLp_L_hr_ = CLp_L_hr__pop*exp(eta.CLp_L_hr_);
  Qper_L_hr_ = Qper_L_hr__pop*exp(eta.Qper_L_hr_);
  Vper_L_ = Vper_L__pop*exp(eta.Vper_L_);
  Vblood_L_ = Vblood_L__pop

  ka_hr_ = ka_hr__pop*exp(eta.ka_hr_);

  fafg = fafg_pop*exp(eta.fafg);

  Hcrit = Hcrit0; # Hematocrit

  Vp_L_ = Vblood_L_*(1 - Hcrit/100);
  Vr_L_ = Vblood_L_*Hcrit/100;
  Qrbc_L_hr_ = 1E5

  fDi = 1 + isDitopic;

  kHbon1_uMhr_ = 2e6*3600*1e-6;
  kHboff1_hr_ = cKD1_uM_*kHbon1_uMhr_;

  kHbon2_uMhr_ = 2e6*3600*1e-6;
```

```
  kHboff2_hr_ = cKD2_uM_*kHbon2_uMhr_;


  CbloodTot_uM_ = (Arbc_umol_ + AHb1rbc_umol_ + 2*AHb2rbc_umol_ +
Ap_umol_)/Vblood_L_;

  Cplasma_uM_ = Ap_umol_/Vp_L_;
  CpFree_uM_ = Cplasma_uM_*fup;

  CplasmaTot_uM_ = (Ap_umol_)/Vp_L_;

  Crbc_uM_ = Arbc_umol_/Vr_L_;                # SN - Conc. of Drug Bound to RBC but unbound
to Hb
  CHb0rbc_uM_ = AHb0rbc_umol_/Vr_L_;          # SN - Conc. of Hb on RBC, not bound to
drug
  CHb1rbc_uM_ = AHb1rbc_umol_/Vr_L_;          # SN - Conc. of Hb, with 1 mol. drug bound
  CHb2rbc_uM_ = AHb2rbc_umol_/Vr_L_;          # SN - Conc. of Hb, with 2 mol. of drug
bound
  CrbcTot_uM_ = (Arbc_umol_ + AHb1rbc_umol_ + 2*AHb2rbc_umol_)/Vr_L_;   # SN - Conc.
of total Hb in blood

  HbCov_Pct_ = 100*(AHb1rbc_umol_ + AHb2rbc_umol_)/(AHb0rbc_umol_ +
AHb1rbc_umol_ + AHb2rbc_umol_);        # SN - Percentage Hb bound in RBC

  Cper_uM_ = Aper_umol_/Vper_L_;

  Cblood.ug.mL. = CbloodTot_uM_*MolWD*1e-3;
  Cplasma.ug.mL. = CplasmaTot_uM_*MolWD*1e-3;
  Crbc.ug.mL. = CrbcTot_uM_*MolWD*1e-3;

  BPR = CbloodTot_uM_/CplasmaTot_uM_;            # Blood to Plasma ratio
  EPR = CrbcTot_uM_/CplasmaTot_uM_;              # Erythrocytes to Plasma ratio
  fuBlood = CpFree_uM_/CbloodTot_uM_;            # fraction unbound in Blood

  # Approximate Hb Coverage calculations - to mimic calculations based on Blood PK
  Cr = (CbloodTot_uM_ - CplasmaTot_uM_*(1-Hcrit/100))/(Hcrit/100);
  HbCov_Cr = 100*Cr/HbTotRBC0_uM_/fDi;
  HbCov_Cb = 100*CbloodTot_uM_/(Hcrit*HbTotRBC0_uM_/100)/fDi;

  kdeathRBC_hr_ = 0;

  AHb0rbc_umol_(0)  = Vr_L_*HbTotRBC0_uM_;      # Total Hb unbound amount in RBCs at
baseline (without Drug)

  Hb.g.dL. = (Hcrit/100)*HbTotRBC0_uM_*64e3*1e-6*1e-1;
```

```
  # ODEs:

  # Drug in Depot
  d/dt(Apo_umol_) = - ka_hr_*Apo_umol_;

  # Amount of free drug in plasma
  d/dt(Ap_umol_) = ka_hr_*fafg*Apo_umol_ -
    CLp_L_hr_*Cplasma_uM_ -
    Qrbc_L_hr_*(CpFree_uM_ - Crbc_uM_) -
    Qper_L_hr_*(Cplasma_uM_ - Cper_uM_) -
     fHemolysis*kdeathRBC_hr_*Arbc_umol_;

  # Amount of drug in peripheral compartment
  d/dt(Aper_umol_) = Qper_L_hr_*(Cplasma_uM_ - Cper_uM_);

  # Amount of drug bound to RBC but not bound to Hb, i.e. free Drug RBC
  d/dt(Arbc_umol_) = Qrbc_L_hr_*(CpFree_uM_ - Crbc_uM_) -
    Vr_L_*(fDi*kHbon1_uMhr_*Crbc_uM_*CHb0rbc_uM_ - kHboff1_hr_*CHb1rbc_uM_) -
    Vr_L_*(kHbon2_uMhr_*Crbc_uM_*CHb1rbc_uM_ - 2*kHboff2_hr_*CHb2rbc_uM_) -
    kdeathRBC_hr_*Arbc_umol_;

  # Amount of Hb on RBC not bound to drug
  d/dt(AHb0rbc_umol_) = kdeathRBC_hr_*Vr_L_*HbTotRBC0_uM_ -
    Vr_L_*(fDi*kHbon1_uMhr_*Crbc_uM_*CHb0rbc_uM_ - kHboff1_hr_*CHb1rbc_uM_) -
    kdeathRBC_hr_*AHb0rbc_umol_;

  # Amount of Hb on RBC with 1 mol of drug bound
  d/dt(AHb1rbc_umol_) = Vr_L_*(fDi*kHbon1_uMhr_*Crbc_uM_*CHb0rbc_uM_ -
kHboff1_hr_*CHb1rbc_uM_) -
    Vr_L_*(kHbon2_uMhr_*Crbc_uM_*CHb1rbc_uM_ - 2*kHboff2_hr_*CHb2rbc_uM_) -
    kdeathRBC_hr_*AHb1rbc_umol_;

  # Amount of Hb on RBC with 2 mol of drug bound
  d/dt(AHb2rbc_umol_) = Vr_L_*(kHbon2_uMhr_*Crbc_uM_*CHb1rbc_uM_ -
2*kHboff2_hr_*CHb2rbc_uM_) -
    kdeathRBC_hr_*AHb2rbc_umol_;
})

summPKPD <- function(PKSim){
  sPKPD <- PKSim %>% group_by(Dose_mg_, Time.hr.) %>%
    summarize(meanCblood.ug.mL. = mean(Cblood.ug.mL., na.rm = T),
              p5Cblood.ug.mL. = quantile(Cblood.ug.mL., probs = 0.05, na.rm = T),
              p50Cblood.ug.mL. = quantile(Cblood.ug.mL., probs = 0.5, na.rm = T),
              p95Cblood.ug.mL. = quantile(Cblood.ug.mL., probs = 0.95, na.rm = T),

              meanCplasma.ug.mL. = mean(Cplasma.ug.mL., na.rm = T),
```

```
            p5Cplasma.ug.mL. = quantile(Cplasma.ug.mL., probs = 0.05, na.rm = T),
            p50Cplasma.ug.mL. = quantile(Cplasma.ug.mL., probs = 0.5, na.rm = T),
            p95Cplasma.ug.mL. = quantile(Cplasma.ug.mL., probs = 0.95, na.rm = T),
            # maxCplasma.ug.mL. = max(Cplasma.ug.ml, na.rm = T),                    ## SN added

            meanCpFree_uM_ = mean(CpFree_uM_, na.rm = T),
            p5CpFree_uM_ = quantile(CpFree_uM_, probs = 0.05, na.rm = T),
            p50CpFree_uM_ = quantile(CpFree_uM_, probs = 0.5, na.rm = T),
            p95CpFree_uM_ = quantile(CpFree_uM_, probs = 0.95, na.rm = T),

            meanHbCov_Pct_ = mean(HbCov_Pct_, na.rm = T),
            p5HbCov_Pct_ = quantile(HbCov_Pct_, probs = 0.05, na.rm = T),
            p50HbCov_Pct_ = quantile(HbCov_Pct_, probs = 0.5, na.rm = T),
            p95HbCov_Pct_ = quantile(HbCov_Pct_, probs = 0.95, na.rm = T),

            meanHcrit = mean(Hcrit, na.rm = T),
            p5Hcrit = quantile(Hcrit, probs = 0.05, na.rm = T),
            p50Hcrit = quantile(Hcrit, probs = 0.5, na.rm = T),
            p95Hcrit = quantile(Hcrit, probs = 0.95, na.rm = T),

            meanHb.g.dL. = mean(Hb.g.dL., na.rm = T),
            p5Hb.g.dL. = quantile(Hb.g.dL., probs = 0.05, na.rm = T),
            p50Hb.g.dL. = quantile(Hb.g.dL., probs = 0.5, na.rm = T),
            p95Hb.g.dL. = quantile(Hb.g.dL., probs = 0.95, na.rm = T))

  return(sPKPD)
}

# Median Body Weight - Not Used
BW_kg_ = 80.7;

fHemolysis = 0;


HbTotRBC0_uM_ = 5e3;

#----------Compound Parameters-------------------------------------
MolWD = 365.37 # Molecular Weight
isDitopic = 1;
fup = 0.162; # Human Plasma Protein Binding Fu

#---------Estimated PK Parameters---------
est <- read.csv(runno, header=TRUE, skip=1, sep="") %>%
  filter(ITERATION == -1E9)

CLp_L_hr__pop = est$THETA1;
```

```
Qper_L_hr__pop = est$THETA2;
Vper_L__pop = est$THETA3;
Vblood_L__pop = est$THETA4;
fafg_pop = est$THETA5;
ka_hr__pop = est$THETA6;
Qrbc_L_hr__pop = 1E5;
cKD1_uM_ = est$THETA11;
cKD2_uM_ = est$THETA12;

omega.CLp_L_hr_ = est$OMEGA.1.1.;
omega.Qper_L_hr_ = est$OMEGA.2.2.;
omega.Vper_L_ = est$OMEGA.3.3.;
omega.ka_hr_ = est$OMEGA.4.4.;
omega.fafg = 0.;
omega.Vblood_L_ = 0.;
omega.Qrbc_L_hr_ = 0.;

Hcrit0 = 43;   # HV hematocrit (%) -
kdeathRBC0_hr_ = 0;

ParsHV <- data.frame(BW_kg_, MolWD, ka_hr__pop, fafg_pop,
                     Vblood_L__pop, CLp_L_hr__pop, Qrbc_L_hr__pop,
                     Vper_L__pop, Qper_L_hr__pop,
                     cKD1_uM_, cKD2_uM_, isDitopic, fup, Hcrit0,
                     HbTotRBC0_uM_, fHemolysis,          # kelHbp_hr_, Vrbc_L_cell_,
                     kdeathRBC0_hr_)

### HV simulation-----------------------------------------------------------
SimType <- "HV BID-PO"
nID = 1000
IDs <- seq(1, nID)
ParsHV$Hcrit0 <- 43    # HV hematocrit
ParsHV1 <- ParsHV[rep(1,times=nID),]

ParsHV1$eta.Vblood_L_ = omega.Vblood_L_*rnorm(nID);
ParsHV1$eta.CLp_L_hr_ = omega.CLp_L_hr_*rnorm(nID);
ParsHV1$eta.Qrbc_L_hr_ = omega.Qrbc_L_hr_*rnorm(nID);
ParsHV1$eta.Vper_L_ = omega.Vper_L_*rnorm(nID);
ParsHV1$eta.Qper_L_hr_ = omega.Qper_L_hr_*rnorm(nID);
ParsHV1$eta.ka_hr_ = omega.ka_hr_*rnorm(nID);
ParsHV1$eta.fafg = omega.fafg*rnorm(nID);
ParsHV1$Hcrit0 <- round(runif(nID, min = 36, max = 50), 2)

Doses_mg_ <- c(1250,2000,2400,2500)    # 1000,1250,1500,1800,
nDose <- 14*2
doseINT_hr_ <- 12
```

```
addls <- nDose-1

Tmeas_hr_ <- seq(0, addls*24, by = 1) %>% unique() %>% sort()

Tm_hr_ <- c(0,0.083,.5,1,2,5,8,11.99,12,c(0.083,.5,1,2,5,8,11.99,12)+12)

Tmeas_hr_ <- unlist(lapply(seq(1,nDose),
                    function(idose){(idose-1)*doseINT_hr_ + Tm_hr_}
))

Tmeass_hr_ <- Tmeas_hr_  %>% unique() %>% sort()

evPO <- et(amt = 1, dosing.to = 1,
        start.time = 0, addl = 1, ii = 12) %>%
  et(Tmeass_hr_)

nD <- length(Doses_mg_)

HV_ind_PK  <- tibble(HbCovAv_Pct_ = as.numeric(),        ## SN - added
                HbCov336_Pct_ = as.numeric(),
                plasma.AUC24ss= as.numeric(),
                plasma.Cmaxss = as.numeric(),
                blood.AUC24ss = as.numeric(),
                blood.Cmaxss  = as.numeric())


for (i in seq(1,nD)){
  indx <- evPO$evid==1
  evPO$amt[indx] = Doses_mg_[i]*1e3/MolWD
  evPO$addl[indx] = addls[1]

  ParsHV1$DoseVal_mg_ = Doses_mg_[i]
  set.seed(999)
  PKSim1 <- rxSolve(simHbSPKmodel1, ParsHV1, evPO, ncores = 8)
  colnames(PKSim1)[1] <- "id"
  colnames(PKSim1)[2] <- "Time.hr."
  PKSim1 <- PKSim1[,c("id","Dose_mg_","Time.hr.","Cblood.ug.mL.",
                  "Cplasma.ug.mL.","HbCov_Pct_","Hcrit",
                  "Hb.g.dL.",'CpFree_uM_')]

  # Calculate average hbcov, hbcov at 336hr and blood and plasma AUC24ss and Cmaxss for
each individual
  hbcov.ind <- subset(PKSim1,Time.hr.>=312 & Time.hr.<=336) %>% group_by(Dose_mg_,id)
%>%
    summarize(HbCovAv_Pct_ = trapz(Time.hr., HbCov_Pct_)/24,
            HbCov336_Pct_ = last(HbCov_Pct_),
```

```
            plasma.AUC24ss = trapz(Time.hr., Cplasma.ug.mL.),
            plasma.Cmaxss = max(Cplasma.ug.mL.),
            blood.AUC24ss = trapz(Time.hr., Cblood.ug.mL.),
            blood.Cmaxss = max(Cblood.ug.mL.))

  HV_ind_PK <- rbind(HV_ind_PK, hbcov.ind)

  # Summary stats SS
  tmp = hbcov.ind %>% group_by(Dose_mg_) %>%
    summarize(HbCovAv_Pct_median = median(hbcov.ind$HbCovAv_Pct_),
            plasma.AUC24ss_median = median(hbcov.ind$plasma.AUC24ss),
            HbCov336_Pct_median = median(hbcov.ind$HbCov336_Pct_),
            plasma.Cmaxss_median = median(hbcov.ind$plasma.Cmaxss),
            blood.AUC24ss_median = median(hbcov.ind$blood.AUC24ss),
            blood.Cmaxss_median = median(hbcov.ind$blood.Cmaxss),

            # plasma unbound NOAEL (1000 mg/kg/day) from 1 month dog toxicity study of
unbound
            # Cmax = 1.5 ug/mL and AUC24 = 22.5 ug?hr/mL (see C4061001 protocol)
            margin.cmax = 1.5/(plasma.Cmaxss_median*0.162),
            margin.auc24 = 22.5/(plasma.AUC24ss_median*0.162))

  sPKSim1 <- summPKPD(PKSim1)
  ## Bind summary stats --
  if (i == 1){
    PKSimAll <- sPKSim1
    av_hbcov.hv = tmp
  }else{
    PKSimAll <- rbind(PKSimAll, sPKSim1)
    av_hbcov.hv = rbind(av_hbcov.hv, tmp)
  }
}

write.csv(HV_ind_PK, './Steady-State-PK-HV.csv', row.names = F, quote = F)      ## SN added
HV_ind_PK_summ <- HV_ind_PK %>% group_by(Dose_mg_) %>% summarise(medianPK =
median(HbCov336_Pct_),
                                                meanPK = mean(HbCov336_Pct_),
                                                sdPK = sd(HbCov336_Pct_),
                                                PK05 = quantile(HbCov336_Pct_, 0.05),
                                                PK95 = quantile(HbCov336_Pct_, 0.95))
write.csv(HV_ind_PK_summ, './HV-Coverage-summary-BID.csv', row.names = F, quote = F)

PKSimAll <- PKSimAll %>% filter(Dose_mg_ != 2400)   ## don't want to plot it

PKSimAll_HV <- PKSimAll
PKSimAll_HV$SimType <- "HV"
```

```
# Plot HV simulated profiles
gCb1 <- ggplot(PKSimAll) +
  geom_line(aes(x = Time.hr./24,
                y = p50Cblood.ug.mL., color = as.factor(Dose_mg_))) +
  scale_x_continuous("Time(days)", breaks = c(0,2,7,14,28)) +
  scale_y_log10("Cblood(ug/mL)", breaks = c(10,100,300,450,600,900),       # 0.1,1,10,100,300
                labels = c(10,100,300,450,600,900)) +                    # 0.1,1,10,100,300
  coord_cartesian(ylim = c(10,900)) +                                  # 0.1,900
  labs(color = "Dose(mg), BID") +
  theme_bwMod1()

gCp1 <- ggplot(PKSimAll) +
  geom_line(aes(x = Time.hr./24,
                y = p50Cplasma.ug.mL., color = as.factor(Dose_mg_))) +
  scale_x_continuous("Time(days)", breaks = c(0,2,7,14,28)) +
  scale_y_log10("Cplasma(ug/mL)", breaks = c(0.1,1,10),          # 0.01,0.1,1,10
                labels = c(0.1,1,3)) +                        # 0.01,0.1,1,10
  coord_cartesian(ylim = c(0.1,3)) +                                   # 0.01,10
  labs(color = "Dose(mg), BID") +
  theme_bwMod1()

gCov1 <- ggplot(PKSimAll) +
  geom_line(aes(x = Time.hr./24,
                y = p50HbCov_Pct_, color = as.factor(Dose_mg_))) +
  scale_x_continuous("Time(days)", breaks = c(0,2,7,14,28)) +
  scale_y_continuous("Hb Coverage(%)", breaks = c(0,10,25,40,60)) +
  coord_cartesian(ylim = c(0,50)) +
  labs(color = "Dose(mg), BID") +
  theme_bwMod1()

gHV <- grid.arrange(gCb1,gCp1,gCov1, ncol = 1,
                    top = paste0("PF-07059013 Predicted PK \n", SimType, ", Hcrit: (",
min(ParsHV1$Hcrit0), " - ", max(ParsHV1$Hcrit0),")"))

gCov3 <- ggplot(PKSimAll_HV) +
  geom_line(aes(x = Time.hr./24, y = p50HbCov_Pct_, color = as.factor(Dose_mg_)), size = 0.5)
+
  geom_ribbon(aes(x = Time.hr./24, ymin = p5HbCov_Pct_, ymax = p95HbCov_Pct_,fill =
as.factor(Dose_mg_)), alpha = 0.3) +
  scale_x_continuous("Time(days)", breaks = c(0,2,7,14,28)) +
  scale_y_continuous("Hb Coverage(%)", breaks = c(0,10,25,40,50)) +
  coord_cartesian(ylim = c(0,50)) +
  facet_wrap(~Dose_mg_, ncol = 1) +
  labs(color = "Dose (mg) BID", fill = "Dose (mg) BID") +
  theme_bwMod1() +
```

```
  ggsave('./Hb-Coverage-with-CI-BID-HV.png', device = 'png', width = 8, height = 6, units = 'in',
dpi = 'retina')

rm(gCov3)
rm(PKSimAll)
### SCD simulation-----------------------------------------------------------------
SimType <- "SCD BID-PO"
nID = 1000
IDs <- seq(1, nID)
ParsSCD = ParsHV
ParsSCD$Hcrit0 <- 27

ParsSCD1 <- ParsSCD[rep(1,times=nID),]

ParsSCD1$eta.Vblood_L_ = omega.Vblood_L_*rnorm(nID);
ParsSCD1$eta.CLp_L_hr_ = omega.CLp_L_hr_*rnorm(nID);
ParsSCD1$eta.Qrbc_L_hr_ = omega.Qrbc_L_hr_*rnorm(nID);
ParsSCD1$eta.Vper_L_ = omega.Vper_L_*rnorm(nID);
ParsSCD1$eta.Qper_L_hr_ = omega.Qper_L_hr_*rnorm(nID);
ParsSCD1$eta.ka_hr_ = omega.ka_hr_*rnorm(nID);
ParsSCD1$eta.fafg = omega.fafg*rnorm(nID);
Hcrit0_SCD <- round(rnorm(n = nID, mean = 27.5, sd = 3.3))    ## mean of 27.5 and sd of 3.3
from HU data
summary(Hcrit0_SCD)
ParsSCD1$Hcrit0 <- Hcrit0_SCD                              #round(runif(nID, min = 17, max = 35))

Doses_mg_ <- c(1250,2000,2250, 2500)        # 1000,1250,1500,1800,
nDose <- 14*2
doseINT_hr_ <- 12                        # BID
addls <- nDose-1

Tmeas_hr_ <- seq(0, addls*24, by = 1) %>% unique() %>% sort()

Tm_hr_ <- c(0,0.083,.5,1,2,5,8,11.99,12,c(0.083,.5,1,2,5,8,11.99,12)+12)

Tmeas_hr_ <- unlist(lapply(seq(1,nDose),
                     function(idose){(idose-1)*doseINT_hr_ + Tm_hr_}
))

Tmeass_hr_ <- Tmeas_hr_  %>% unique() %>% sort()

evPO <- et(amt = 1, dosing.to = 1,
         start.time = 0, addl = 1, ii = 12) %>%
  et(Tmeass_hr_)

nD <- length(Doses_mg_)
```

```
SCD_ind_PK <- tibble(HbCovAv_Pct_ = as.numeric(),        ## SN - added
                     HbCov336_Pct_ = as.numeric(),
                     plasma.AUC24ss= as.numeric(),
                     plasma.Cmaxss = as.numeric(),
                     blood.AUC24ss = as.numeric(),
                     blood.Cmaxss  = as.numeric())

for (i in seq(1,nD)){
  indx <- evPO$evid==1
  evPO$amt[indx] = Doses_mg_[i]*1e3/MolWD
  evPO$addl[indx] = addls[1]

  ParsSCD1$DoseVal_mg_ = Doses_mg_[i]
  set.seed(999)
  PKSim1 <- rxSolve(simHbSPKmodel1, ParsSCD1, evPO, ncores = 8)
  colnames(PKSim1)[1] <- "id"
  colnames(PKSim1)[2] <- "Time.hr."
  PKSim1 <- PKSim1[,c("id","Dose_mg_","Time.hr.","Cblood.ug.mL.",
                     "Cplasma.ug.mL.","HbCov_Pct_","Hcrit",
                     "Hb.g.dL.",'CpFree_uM_')]

  # Calculate average hbcov, hbcov at 336hr and blood and plasma AUC24ss and Cmaxss for
each individual
  hbcov.ind <- subset(PKSim1,Time.hr.>=312 & Time.hr.<=336) %>% group_by(Dose_mg_,id)
%>%
    summarize(HbCovAv_Pct_ = trapz(Time.hr., HbCov_Pct_)/24,
              HbCov336_Pct_ = last(HbCov_Pct_),
              plasma.AUC24ss = trapz(Time.hr., Cplasma.ug.mL.),
              plasma.Cmaxss = max(Cplasma.ug.mL.),
              blood.AUC24ss = trapz(Time.hr., Cblood.ug.mL.),
              blood.Cmaxss = max(Cblood.ug.mL.))

  SCD_ind_PK = rbind(SCD_ind_PK, hbcov.ind)              ## SN - added

  # Summary stats SS
  tmp = hbcov.ind %>% group_by(Dose_mg_) %>%
    summarize(HbCovAv_Pct_median = median(hbcov.ind$HbCovAv_Pct_),
              plasma.AUC24ss_median = median(hbcov.ind$plasma.AUC24ss),
              HbCov336_Pct_median = median(hbcov.ind$HbCov336_Pct_),
              plasma.Cmaxss_median = median(hbcov.ind$plasma.Cmaxss),
              blood.AUC24ss_median = median(hbcov.ind$blood.AUC24ss),
              blood.Cmaxss_median = median(hbcov.ind$blood.Cmaxss),

              # plasma unbound NOAEL (1000 mg/kg/day) from 1 month dog toxicity study of
unbound
```

```
        # Cmax = 1.5 ?g/mL and AUC24 = 22.5 ?g?hr/mL (see C4061001 protocol)
        margin.cmax = 1.5/(plasma.Cmaxss_median*0.162),
        margin.auc24 = 22.5/(plasma.AUC24ss_median*0.162))

  sPKSim1 <- summPKPD(PKSim1)

  if (i == 1){
    PKSimAll <- sPKSim1
    av_hbcov.scd = tmp
  }else{
    PKSimAll <- rbind(PKSimAll, sPKSim1)
    av_hbcov.scd = rbind(av_hbcov.scd, tmp)
  }
}

write.csv(SCD_ind_PK, './Steady-State-PK-SCD.csv', row.names = F, quote = F)      ## SN
added
SCD_ind_PK_summ <- SCD_ind_PK %>% group_by(Dose_mg_) %>% summarise(medianPK
= median(HbCov336_Pct_),
                                                meanPK = mean(HbCov336_Pct_),
                                                sdPK = sd(HbCov336_Pct_),
                                                PK05 = quantile(HbCov336_Pct_, 0.05),
                                                PK95 = quantile(HbCov336_Pct_, 0.95))
write.csv(SCD_ind_PK_summ, './SCD-Coverage-summary-BID.csv', row.names = F, quote = F)

PKSimAll_SCD <- PKSimAll
PKSimAll_SCD$SimType <- "SCD"
PKSimAll_SCD$nDose <- nDose
PKSimAll_SCD$doseINT_hr_ <- doseINT_hr_

# HbCov336_Pct_median at 1500 mg BID and 2000 mg BID for SCD
hbav2500 = as.numeric(av_hbcov.scd[av_hbcov.scd$Dose_mg_==1250,
'HbCov336_Pct_median'])
hbav2500
hbav4000 = as.numeric(av_hbcov.scd[av_hbcov.scd$Dose_mg_==2000,
'HbCov336_Pct_median'])
hbav4000
hbav4500 = as.numeric(av_hbcov.scd[av_hbcov.scd$Dose_mg_==2250,
'HbCov336_Pct_median'])
hbav4500
hbav5000 = as.numeric(av_hbcov.scd[av_hbcov.scd$Dose_mg_==2500,
'HbCov336_Pct_median'])
hbav5000

# Summary stats of av_Hbcov
av_hbcov.hv$SimType <- "HV"
```

```
av_hbcov.scd$SimType <- "SCD"
av_hbcov.sum = rbind(av_hbcov.hv,av_hbcov.scd)
write.csv(av_hbcov.sum, paste0(projRun,"-av_hbcov.sum.csv"), row.names = F)

# Plot SCD simulated profiles
gCb2 <- ggplot(PKSimAll) +
  geom_line(aes(x = Time.hr./24,
            y = p50Cblood.ug.mL., color = as.factor(Dose_mg_))) +
  scale_x_continuous("Time(days)", breaks = c(0,2,7,14,28)) +
  scale_y_log10("Cblood(ug/mL)", breaks = c(10,100,300),
            labels = c(10,100,300)) +
  coord_cartesian(ylim = c(10,450)) +
  labs(color = "Dose(mg), BID") +
  theme_bwMod1() +
  ggsave('./Blood-concentrations-SCD.png', device = 'png', width = 8, height = 6, dpi = 'retina')

gCp2 <- ggplot(PKSimAll) +
  geom_line(aes(x = Time.hr./24,
            y = p50Cplasma.ug.mL., color = as.factor(Dose_mg_))) +
  scale_x_continuous("Time(days)", breaks = c(0,2,7,14,28)) +
  scale_y_log10("Cplasma(ug/mL)", breaks = c(0.1,1,2.5),
            labels = c(0.1,1,2.5)) +
  coord_cartesian(ylim = c(0.1,2.5)) +
  labs(color = "Dose(mg), BID") +
  theme_bwMod1() +
  ggsave('./Plasma-concentrations-SCD.png', device = 'png', width = 8, height = 6, dpi = 'retina')

# Plot predicted Hb coverage with dashed lines for HbCovAv_Pct_median at 3000 mg and 3500
mg for SCD
n = 7
cols = gg_color_hue(n)

gCov2 <- ggplot(PKSimAll) +
  geom_line(aes(x = Time.hr./24,
            y = p50HbCov_Pct_, color = as.factor(Dose_mg_))) +
  scale_x_continuous("Time(days)", breaks = c(0,2,7,14,28)) +
  scale_y_continuous("Hb Coverage(%)", breaks = c(0,10,25,40,50)) +
  coord_cartesian(ylim = c(0,50)) +
  labs(color = "Dose (mg), BID") +
  theme_bwMod1() +
  ggsave('./Hb-coverage-SCD.png', device = 'png', width = 8, height = 6, dpi = 'retina')

gSCD <- grid.arrange(gCb2,gCp2,gCov2, ncol = 1,
                 top = paste0("PF-07059013 Predicted PK \n", SimType, ", Hcrit: (",
min(ParsSCD1$Hcrit0), " - ", max(ParsSCD1$Hcrit0),")"))
```

```
write.csv(PKSimAll_HV, paste0(projRun,"-HV PO-BID PK Pred.csv"), row.names = F)
write.csv(PKSimAll_SCD, paste0(projRun,"-SCD PO-BID PK Pred.csv"), row.names = F)

pdf(paste0(projRun,"-MultiDose PKModel Sim-1.pdf"),
   width = 12, height = 8, useDingbats = F)
grid.arrange(gHV, gSCD, nrow = 1)

dev.off()

pdf(paste0(projRun,"-MultiDose Hbcov Sim-1.pdf"),
   width = 8, height = 8, useDingbats = F)

#hv.title =  paste0("Predicted Hb coverage - HV Hcrit: (", min(ParsHV1$Hcrit0), " - ",
max(ParsHV1$Hcrit0), ")")
#scd.title =  paste0("Predicted Hb coverage - SCD Hcrit: (", min(ParsSCD1$Hcrit0), " - ",
max(ParsSCD1$Hcrit0), ")")
hv.title = paste0("Predicted Hb coverage - HV")
scd.title = paste0("Predicted Hb coverage - SCD")
grid.arrange(gCov1 + ggtitle(hv.title),
          gCov2 + ggtitle(scd.title) +
            geom_hline(aes(yintercept = hbav2500),color='black', linetype = 2, alpha = 0.4) +
            geom_hline(aes(yintercept = hbav4000),color='black', linetype = 2, alpha = 0.4) +
            geom_hline(aes(yintercept = hbav4500),color='black', linetype = 2, alpha = 0.4)) # +
            # geom_hline(aes(yintercept = hbav5000),color='black', linetype = 2, alpha = 0.4)) #+
            # scale_linetype_manual(name = "Median SS\n average Hb\n coverage"), values =
rep(2,4))
          #   scale_linetype_manual(name = "Median SS\n average Hb\n coverage", values =
rep(2,3),
          #                    guide = guide_legend(override.aes = list(color = c(cols[1], cols[2],
cols[3], cols[4])))),
          # nrow = 2)

dev.off()

## saving as png
png(paste0(projRun,"-MultiDose Hbcov Sim-1.png"), width = 8, height = 8, units = 'in', res =
320)

#hv.title =  paste0("Predicted Hb coverage - HV Hcrit: (", min(ParsHV1$Hcrit0), " - ",
max(ParsHV1$Hcrit0), ")")
#scd.title =  paste0("Predicted Hb coverage - SCD Hcrit: (", min(ParsSCD1$Hcrit0), " - ",
max(ParsSCD1$Hcrit0), ")")
hv.title = paste0("Predicted Hb coverage - HV")
scd.title = paste0("Predicted Hb coverage - SCD")
grid.arrange(gCov1 + ggtitle(hv.title),
          gCov2 + ggtitle(scd.title) +
```

```
            geom_hline(aes(yintercept = hbav2500),color='black', linetype = 2, alpha = 0.4) +
            geom_hline(aes(yintercept = hbav4000),color='black', linetype = 2, alpha = 0.4) +
            geom_hline(aes(yintercept = hbav4500),color='black', linetype = 2, alpha = 0.4)) # +
            # geom_hline(aes(yintercept = hbav5000),color='black', linetype = 2, alpha = 0.4)) #+
            # scale_linetype_manual(name = "Median SS\n average Hb\n coverage"), values =
rep(2,4))
          #   scale_linetype_manual(name = "Median SS\n average Hb\n coverage", values =
rep(2,3),
          #                    guide = guide_legend(override.aes = list(color = c(cols[1], cols[2],
cols[3], cols[4]))))),
          # nrow = 2)

dev.off()

gCov3 <- ggplot(PKSimAll_SCD) +
  geom_line(aes(x = Time.hr./24, y = p50HbCov_Pct_, color = as.factor(Dose_mg_)), size = 0.5)
+
  geom_ribbon(aes(x = Time.hr./24, ymin = p5HbCov_Pct_, ymax = p95HbCov_Pct_,fill =
as.factor(Dose_mg_)), alpha = 0.3) +
  scale_x_continuous("Time(days)", breaks = c(0,2,7,14,28)) +
  scale_y_continuous("Hb Coverage(%)", breaks = c(0,10,25,40,60)) +
  coord_cartesian(ylim = c(0,70)) +
  facet_wrap(~Dose_mg_) +
  labs(color = "Dose (mg) BID", fill = "Dose (mg) BID") +
  theme_bwMod1() +
  ggsave('./Hb-Coverage-with-CI-BID.png', device = 'png', width = 8, height = 6, units = 'in', dpi
= 'retina')
```

Code for Figure 3 C

```
### Linear Model between Delta p20 and Hb Coverage
## plot the HbCov with p20 and deltap20 data

library(dplyr)
library(ggplot2)
library(broom)

NMdata <- read.csv('./C4061001-21SEP-v2.csv', header = T, stringsAsFactors = F)
sdtab55 <- read.csv('./sdtab55', header = T, sep = "", stringsAsFactors = F)

deltap20 <- NMdata %>% group_by(ID) %>% filter(FLAG == 4) %>%
  mutate(delp20 = as.numeric(DV) - first(as.numeric(DV)),
        DV = as.numeric(DV), BLQ = as.numeric(BLQ)) %>%
  rename(TIME = TAFD)

combined <- dplyr::full_join(sdtab55, deltap20,
```

```
                    by = c("ID", "DOSE", "TRT", "TIME", "COH", "PERD",
                         "FLAG", "HCT", "DV", "MDV", "BLQ", "EVID"))

combined <- combined %>% arrange(ID, TIME, FLAG)
write.csv(combined, './all-records-combined-p20-Coverage.csv', row.names = F, quote = F)

##########
sdtab55$DV <- as.numeric(sdtab55$DV)
sdtab55s <- sdtab55 %>% select(ID, TRT, TIME, EVID, MDV, BLQ, FLAG, DV, IPRED,
HbCov)

deltap20 <- NMdata %>% group_by(ID) %>% filter(FLAG == 4) %>%
 mutate(delp20 = as.numeric(DV) - first(as.numeric(DV))) %>%
 select(ID, TRT, TIME = TAFD, FLAG, EVID, MDV, BLQ, DV, delp20)
deltap20$DV <- as.numeric(deltap20$DV)
deltap20$BLQ <- as.numeric(deltap20$BLQ)

test <- dplyr::full_join(sdtab55s, deltap20, by =
c('ID','TRT','TIME','EVID','MDV','FLAG','BLQ','DV'))
test <- test %>% arrange(ID, TIME, FLAG)
write.csv(test, './all-records-combined-p20-main-columns.csv', row.names = F, quote = F)

## split and combine the dataset to get HbCov and delp20 merged by ID and TIME
df1 <- test %>%  filter(FLAG == 1) %>% select(ID, TIME,HbCov)
df2 <- test %>% filter(FLAG == 4) %>% select(ID, TIME, delp20)

test1 <- dplyr::full_join(df1, df2, by = c("ID","TIME")) %>% arrange(ID, TIME)
sel <- test1$TIME == 0.0
test1[sel, 'HbCov'] <- 0

placebo_ID <- NMdata[NMdata$TRTG == 'Placebo_MD', ]$ID %>% unique()
test1[test1$ID %in% placebo_ID, 'HbCov'] <- 0
test1$delp20 <- -1 * test1$delp20

## set HbCov to zero when BLQ is 1 for plasma or blood PK
sel <- NMdata$FLAG %in% c(1,2) & NMdata$BLQ == 1
BLQdf <- NMdata[sel, c('ID', 'TAFD', 'BLQ')]
BLQdf <- BLQdf %>% rename(TIME = TAFD) %>% distinct()

test1 <- dplyr::full_join(test1, BLQdf, by = c("ID", "TIME"))
test1 <- test1[!is.na(test1$delp20), ]
sel <- test1$BLQ == 1 & !is.na(test1$BLQ)
test1[sel, 'HbCov'] <- 0

test1 <- test1 %>% select(-BLQ)      ## 398 total rows
```

```
## also set HbCov when DOSE column in NMdata is 0
sel <- NMdata$FLAG == 4 & NMdata$DOSE == 0
DOSEdf <- NMdata[sel, c('ID', 'TAFD', 'DOSE')]
DOSEdf <- DOSEdf %>% rename(TIME = TAFD) %>% distinct()

test1 <- dplyr::left_join(test1, DOSEdf, by = c("ID", "TIME"))
sel <- test1$DOSE == 0 & !is.na(test1$DOSE)
test1[sel, 'HbCov'] <- 0
test1 <- test1 %>% select(-DOSE)   ## 398 rows

sel <- complete.cases(test1)
test2 <- test1[sel, ]              ## 388 rows

write.csv(test2, './Hbcov-delp20-for-modeling.csv', row.names = F, quote = F)

p1 <- ggplot(test2, aes(x = HbCov, y = delp20)) +
 geom_point() +
 geom_smooth(method = 'lm', formula = 'y ~ x + 0', se = F) +
 ylab('delta p20') + xlab("HbCov (%)") +
 theme_bw() +
 ggsave('./deltap20-with-HbCoverage.png', device = 'png', width = 8, height = 6, dpi = 'retina')
p1


## linear model, no intercept -----------------------------------------
lin_model <- lm(delp20 ~ HbCov + 0, data = test2)
sink("linear-model-no-intercept.txt")
summary(lin_model)
sink()
pdf('linear-model-no-intercept.pdf', width = 8, height = 6)
plot(lin_model)
dev.off()

## simulation
set.seed(12345)
tv_slope <- tidy(lin_model)$estimate   ## typical value
sd_slope <- sqrt(nrow(test2)) * tidy(lin_model)$std.error  ## sd = sqrt(n) * se
rse <- glance(lin_model)$sigma

## simulate with assumed 25% coverage
N <- 11500
slope <- tv_slope + rnorm(n = N, mean = 0, sd = sd_slope)

## load the mean and sd coverage from HV 2000 mg BID summary data
HV_cov_summary <- read.csv('./HV-Coverage-summary-BID.csv', header = T, stringsAsFactors
= F)
```

```
## use the dose which gives closest AUC/Cmax compared to 4000 mg TDD
mean_Coverage <- HV_cov_summary[HV_cov_summary$Dose_mg_ == 2400, 'meanPK']
sd_Coverage <- HV_cov_summary[HV_cov_summary$Dose_mg_ == 2400, 'sdPK']
y05_Coverage <- HV_cov_summary[HV_cov_summary$Dose_mg_ == 2400, 'PK05']
y95_Coverage <- HV_cov_summary[HV_cov_summary$Dose_mg_ == 2400, 'PK95']

Coverage <- rnorm(n = N, mean = mean_Coverage, sd = sd_Coverage)
summary(Coverage)
sim_df <- data.frame("slope" = slope, "coverage" = Coverage)
sim_df <- sim_df %>% filter(coverage >= y05_Coverage, coverage <= y95_Coverage)
summary(sim_df$coverage)
y <- sim_df$slope * sim_df$coverage + rnorm(n = nrow(sim_df), mean = 0, sd = sqrt(rse))

## for 90% Prediction Interval
y90PI <- quantile(y, probs = c(0.05, 0.50, 0.95))
y90PI_df <- data.frame("p05" = y90PI[1], "p50" = y90PI[2], "p95" = y90PI[3])
write.csv(y90PI_df, './PI90-no-intercept.csv', row.names = F, quote = F)

## linear model, no intercept with ID as group
lin_model_ID <- lm(delp20 ~ HbCov + 0 + (1 | ID), data = test2)
sink("linear-model-no-intercept-ID.txt")
summary(lin_model_ID)
sink()
pdf('linear-model-no-intercept-ID.pdf', width = 8, height = 6)
plot(lin_model_ID)
dev.off()


## linear model, with intercept -----------------------------------------
lin_model <- lm(-delp20 ~ HbCov, data = test2)
sink("linear-model-with-intercept.txt")
summary(lin_model)
sink()
pdf('linear-model-with-intercept.pdf', width = 8, height = 6)
plot(lin_model)
dev.off()


## linear model, with intercept with ID as group
lin_model_ID <- lm(-delp20 ~ HbCov + (1 | ID), data = test2)
sink("linear-model-with-intercept-ID.txt")
summary(lin_model_ID)
sink()
pdf('linear-model-with-intercept-ID.pdf', width = 8, height = 6)
plot(lin_model_ID)
dev.off()
```

Code for Figure 4

```
function ydot = HbS_ODE(t,y,pars,Dose)

%% variables
yy = num2cell(y);

[Apo_umol_, Ap_umol_, AHb0p_umol_, AHb1p_umol_, AHb2p_umol_, ...
Aper_umol_, Arbc_umol_, AHb0rbc_umol_, AHb1rbc_umol_, AHb2rbc_umol_,...
EPO_U_mL_ , NCFUe_e12cells_, Nret_e12cells_, Nrbc_e12cells_ ] = deal(yy{:});

%% parameters
Dose_mg_ = Dose;

p = num2cell(pars);
[MolWD, ka_hr_, fbio, Vblood_L_, CLp_L_hr_, Qrbc_L_hr_, ...
Vper_L_, Qper_L_hr_, cKD1_uM_, cKD2_uM_, fup, Hcrit0, ...
HbTotRBC0_uM_, fHemolysis, kelHbp_hr_, ...
kdeathRBC0_hr_, Imaxkd, KCov_Pct_, nd, ...
vEPOmax_U_mLhr_, Krbc_e12cells_, n, kelEPO_hr_, ...
Vrbc_L_cell_, vrmin_e12cells_hr_, EmaxEPO, Kepo_U_mL_, m, ...
kCFUeRet_hr_, kret_hr_, kHbon1_uMhr_] = deal(p{:});

isDitopic = 1;

%%
% Retic Fraction
RetFrac = Nret_e12cells_/Nrbc_e12cells_;
% Total RBC
NrbcAll_e12cells_ = Nret_e12cells_ + Nrbc_e12cells_;
% Hematocrit
Hcrit = 100*max(NrbcAll_e12cells_*1e12*Vrbc_L_cell_/Vblood_L_,0);

% Plasma and RBC total volume
Vp_L_ = Vblood_L_*(1 - Hcrit/100);
Vr_L_ = Vblood_L_*Hcrit/100;
```

```
% ditopic binding or not
fDi = 1 + isDitopic;

% drug binding rate constants
kHboff1_hr_ = cKD1_uM_*kHbon1_uMhr_;

kHbon2_uMhr_ = isDitopic*kHbon1_uMhr_;
kHboff2_hr_ = cKD2_uM_*kHbon2_uMhr_;

%% concentrations
CbloodTot_uM_ = (Arbc_umol_ + AHb1rbc_umol_ + 2*AHb2rbc_umol_ + ...
                Ap_umol_ + AHb1p_umol_ + 2*AHb2p_umol_)/Vblood_L_;

Cplasma_uM_ = Ap_umol_/Vp_L_;
CpFree_uM_ = Cplasma_uM_*fup;

CHb0p_uM_ = AHb0p_umol_/Vp_L_;
CHb1p_uM_ = AHb1p_umol_/Vp_L_;
CHb2p_uM_ = AHb2p_umol_/Vp_L_;
CplasmaTot_uM_ = (Ap_umol_ + AHb1p_umol_ + 2*AHb2p_umol_)/Vp_L_;

Crbc_uM_ = Arbc_umol_/Vr_L_;
CHb0rbc_uM_ = AHb0rbc_umol_/Vr_L_;
CHb1rbc_uM_ = AHb1rbc_umol_/Vr_L_;
CHb2rbc_uM_ = AHb2rbc_umol_/Vr_L_;
CrbcTot_uM_ = (Arbc_umol_ + AHb1rbc_umol_ + 2*AHb2rbc_umol_)/Vr_L_;

Cper_uM_ = Aper_umol_/Vper_L_;

Cblood_ug_mL_ = CbloodTot_uM_*MolWD*1e-3;
Cplasma_ug_mL_ = CplasmaTot_uM_*MolWD*1e-3;
Crbc_ug_mL_ = CrbcTot_uM_*MolWD*1e-3;



BPR = CbloodTot_uM_/CplasmaTot_uM_;
EPR = CrbcTot_uM_/CplasmaTot_uM_;
fuBlood = CpFree_uM_/CbloodTot_uM_;
```

```
%% Hb Coverage by drug
if (AHb0rbc_umol_ + AHb1rbc_umol_ + AHb2rbc_umol_)==0
   HbCov_Pct_=0;
else
   HbCov_Pct_ = 100*(AHb1rbc_umol_ + AHb2rbc_umol_)/(AHb0rbc_umol_ +
AHb1rbc_umol_ + AHb2rbc_umol_);
end
Cr = (CbloodTot_uM_ - CplasmaTot_uM_*(1-Hcrit/100))/(Hcrit/100);
HbCov_Cr = 100*Cr/HbTotRBC0_uM_/fDi;
HbCov_Cb = 100*CbloodTot_uM_/(Hcrit*HbTotRBC0_uM_/100)/fDi;

% O2 saturation p50/p20 -- not used
dp50_mmHg_ = 0.12*HbCov_Pct_;
dp20_mmHg_ = 0.78*HbCov_Pct_;

% EPO production rate - dependent on total RBC
vEPO_U_mLhr_ = vEPOmax_U_mLhr_*Krbc_e12cells_^n/(Krbc_e12cells_^n +
NrbcAll_e12cells_^n);
% progenitor cell birth rate - dependent on EPO
kbirthCFUe_e12cells_hr_ = vrmin_e12cells_hr_*(1 + ...
                    EmaxEPO*EPO_U_mL_^m/(Kepo_U_mL_^m + EPO_U_mL_^m));
% RBC death rate - dependent on Hb Cov by drug
kdeathRBC_hr_ = kdeathRBC0_hr_*(1 - Imaxkd*HbCov_Pct_^nd/(KCov_Pct_^nd +
HbCov_Pct_^nd));


%% ODEs
% oral compartment
dApo_umol_ = - ka_hr_*Apo_umol_;

% plasma
dAp_umol_ = ka_hr_*fbio*Apo_umol_ - ...
            CLp_L_hr_*Cplasma_uM_ -...
            Qper_L_hr_*(Cplasma_uM_ - Cper_uM_) - ...
            Qrbc_L_hr_*(CpFree_uM_ - Crbc_uM_) - ...
            Vp_L_*(fDi*kHbon1_uMhr_*CpFree_uM_*CHb0p_uM_ -
kHboff1_hr_*CHb1p_uM_) - ...
```

```
            Vp_L_*(kHbon2_uMhr_*CpFree_uM_*CHb1p_uM_ -
2*kHboff2_hr_*CHb2p_uM_) + ...
            fHemolysis*kdeathRBC_hr_*(Nrbc_e12cells_/NrbcAll_e12cells_)*Arbc_umol_;

% Hb0p, Hb1p, Hb2p -- Hb in plasma; these variables will be zero when
% fHemolysis = 0
dAHb0p_umol_ = - Vp_L_*(fDi*kHbon1_uMhr_*CpFree_uM_*CHb0p_uM_ -
kHboff1_hr_*CHb1p_uM_) + ...

fHemolysis*kdeathRBC_hr_*(Nrbc_e12cells_/NrbcAll_e12cells_)*AHb0rbc_umol_ - ...
              kelHbp_hr_*AHb0p_umol_;

dAHb1p_umol_ = Vp_L_*(fDi*kHbon1_uMhr_*CpFree_uM_*CHb0p_uM_ -
kHboff1_hr_*CHb1p_uM_) - ...
              Vp_L_*(kHbon2_uMhr_*CpFree_uM_*CHb1p_uM_ -
2*kHboff2_hr_*CHb2p_uM_) -...

fHemolysis*kdeathRBC_hr_*(Nrbc_e12cells_/NrbcAll_e12cells_)*AHb1rbc_umol_ - ...
              kelHbp_hr_*AHb1p_umol_;

dAHb2p_umol_ = Vp_L_*(kHbon2_uMhr_*CpFree_uM_*CHb1p_uM_ -
2*kHboff2_hr_*CHb2p_uM_) - ...

fHemolysis*kdeathRBC_hr_*(Nrbc_e12cells_/NrbcAll_e12cells_)*AHb2rbc_umol_ - ...
              kelHbp_hr_*AHb2p_umol_;

% peripheral compartment
% peripheral compartment will be 0 for GBT
dAper_umol_ = Qper_L_hr_*(Cplasma_uM_ - Cper_uM_);


% RBC compartment
dArbc_umol_ = Qrbc_L_hr_*(CpFree_uM_ - Crbc_uM_) - ...
              Vr_L_*(fDi*kHbon1_uMhr_*Crbc_uM_*CHb0rbc_uM_ -
kHboff1_hr_*CHb1rbc_uM_) -...
              Vr_L_*(kHbon2_uMhr_*Crbc_uM_*CHb1rbc_uM_ -
2*kHboff2_hr_*CHb2rbc_uM_) - ...
              kdeathRBC_hr_*(Nrbc_e12cells_/NrbcAll_e12cells_)*Arbc_umol_;
```

```
dAHb0rbc_umol_ = kCFUeRet_hr_*NCFUe_e12cells_*1e12*Vrbc_L_cell_*HbTotRBC0_uM_
- ...
                Vr_L_*(fDi*kHbon1_uMhr_*Crbc_uM_*CHb0rbc_uM_ -
kHboff1_hr_*CHb1rbc_uM_) - ...
                kdeathRBC_hr_*(Nrbc_e12cells_/NrbcAll_e12cells_)*AHb0rbc_umol_;

dAHb1rbc_umol_ = Vr_L_*(fDi*kHbon1_uMhr_*Crbc_uM_*CHb0rbc_uM_ -
kHboff1_hr_*CHb1rbc_uM_) - ...
                Vr_L_*(kHbon2_uMhr_*Crbc_uM_*CHb1rbc_uM_ -
2*kHboff2_hr_*CHb2rbc_uM_) -...
                kdeathRBC_hr_*(Nrbc_e12cells_/NrbcAll_e12cells_)*AHb1rbc_umol_;

dAHb2rbc_umol_ = Vr_L_*(kHbon2_uMhr_*Crbc_uM_*CHb1rbc_uM_ -
2*kHboff2_hr_*CHb2rbc_uM_) - ...
                kdeathRBC_hr_*(Nrbc_e12cells_/NrbcAll_e12cells_)*AHb2rbc_umol_;

dEPO_U_mL_ = vEPO_U_mLhr_ - kelEPO_hr_*EPO_U_mL_;
dNCFUe_e12cells_ = kbirthCFUe_e12cells_hr_ - kCFUeRet_hr_*NCFUe_e12cells_;
dNret_e12cells_ = kCFUeRet_hr_*NCFUe_e12cells_ - kret_hr_*Nret_e12cells_;
dNrbc_e12cells_ = kret_hr_*Nret_e12cells_ - kdeathRBC_hr_*Nrbc_e12cells_;

ydot = [dApo_umol_; dAp_umol_; dAHb0p_umol_; dAHb1p_umol_; dAHb2p_umol_; ...
   dAper_umol_; dArbc_umol_; dAHb0rbc_umol_; dAHb1rbc_umol_; dAHb2rbc_umol_;...
   dEPO_U_mL_ ; dNCFUe_e12cells_; dNret_e12cells_;dNrbc_e12cells_ ];

clear, close all
% script to generate figures

% PK parameters (from Table S2) -- these are all fixed across Vpop
% Vblood_L_ = 5;
% % ka_hr_ = 0.666; % fasted
% ka_hr_ = 0.4376;  % fed condition -- USED for simulations in manuscript
% % fbio = 0.608;   % fasted
% fbio = 0.7679;    % fed  -- USED for simulations in manuscript
% CLp_L_hr_ = 85;
% Qper_L_hr_ = 132;
% Vper_L_ = 1200;
% cKD1_uM_ = 68.6;
% cKD2_uM_ = 0.00992;
```

```
% MolWD = 365.37;
% fup = 0.162;
% Qrbc_L_hr_ = 1e5;
% PK parameters all loaded into parset parameter array

%% plot RBC half-life - Fig 4A
% load vpop parameters
parset = table2array(readtable('vpop.csv','ReadVariableNames', false));

HbCov = [0:1:100];
RBC_life_Vec = zeros(length(parset),length(HbCov));

for kk =1:length(parset)

% set parameters
pars = parset(kk,:);
p = num2cell(pars);

[MolWD, ka_hr_, fbio, Vblood_L_, CLp_L_hr_, Qrbc_L_hr_, ...
Vper_L_, Qper_L_hr_, cKD1_uM_, cKD2_uM_, fup, ~, ...
HbTotRBC0_uM_, fHemolysis, kelHbp_hr_, ...
kdeathRBC0_hr_, Imaxkd, KCov_Pct_, nd, ...
vEPOmax_U_mLhr_, Krbc_e12cells_, n, kelEPO_hr_, ...
Vrbc_L_cell_, vrmin_e12cells_hr_, EmaxEPO, Kepo_U_mL_, m, ...
kCFUeRet_hr_, kret_hr_, kHbon1_uMhr_] = deal(p{:});

   for i = 1:length(HbCov)
      HbCov_Pct_ = HbCov(i);
      kdeathRBC_hr_(i) = kdeathRBC0_hr_*(1 - Imaxkd*HbCov_Pct_^nd/(KCov_Pct_^nd +
HbCov_Pct_^nd));
   end
   RBC_life = log(2)./kdeathRBC_hr_/24;

   RBC_life_Vec(kk,:) = RBC_life;
end
median_RBC_life = median(RBC_life_Vec);
RBC_life_5 = quantile(RBC_life_Vec,0.05);
RBC_life_95 = quantile(RBC_life_Vec,0.95);
```

```
f1 = figure('Name','RBC Lifetime','Units','Inches','Position',[1 1 8 9]);
patch([HbCov,fliplr(HbCov)],[RBC_life_95,fliplr(RBC_life_5)],...
'b','facealpha',0.3,'edgecolor','none'), hold on
plot(HbCov,median_RBC_life,'color',[0.6350 0.0780 0.1840],'linewidth',2)
ylim([0 120]), hold on
ylabel('RBC Half Life (Days)')
xlabel('Hb Occupancy %')
set(gca,'fontsize',24)
legend({'90% Interval','Median'},'Location','northwest')
saveas(f1,'rbc_lifetime_var.png')

%% plot time course over time - Fig 4B
clear

% load vpop parameters
parset = table2array(readtable('vpop.csv','ReadVariableNames', false));
Nsample = length(parset);   % number of virtual population

% dosing info
Dose_mg = 1250;        % BID Dosing
nDose = 8*7*2;         % number of dose BID
addlDay = 10*7;
Nweeks = (nDose+addlDay)/7;
dt = 2;           % hour
tlen = nDose*(12/dt+1)+(addlDay*24/dt+1);
tsim = zeros(Nsample,tlen);

% initialize model output
dHb_g_dL_ind = zeros(Nsample,tlen);
HbCov_Pct_ind = zeros(Nsample,tlen);

% loop over individual
parfor ii = 1:Nsample
    % load pars
    pars = parset(ii,:);
    p = num2cell(pars);

    [MolWD, ka_hr_, fbio, Vblood_L_, CLp_L_hr_, Qrbc_L_hr_, ...
```

```
Vper_L_, Qper_L_hr_, cKD1_uM_, cKD2_uM_, fup, Hcrit0, ...
HbTotRBC0_uM_, fHemolysis, kelHbp_hr_, ...
kdeathRBC0_hr_, Imaxkd, KCov_Pct_, nd, ...
vEPOmax_U_mLhr_, Krbc_e12cells_, n, kelEPO_hr_, ...
Vrbc_L_cell_, vrmin_e12cells_hr_, EmaxEPO, Kepo_U_mL_, m, ...
kCFUeRet_hr_, kret_hr_, kHbon1_uMhr_] = deal(p{:});

% set IC
[EPO0_U_mL_, NCFUe0_e12cells_,Nret0_e12cells_, Nrbc0_e12cells_, ...
        NrbcAll0_e12cells_,RetFrac0, Hcrit0,AHb0rbc0]  = setIC_Ind(pars);

y0 = zeros(14,1);
y0(8) = AHb0rbc0; y0(11) = EPO0_U_mL_; y0(12) = NCFUe0_e12cells_;
y0(13) = Nret0_e12cells_;  y0(14) = Nrbc0_e12cells_;
y00 = y0;
Dose_umol_ = Dose_mg*1e3/MolWD;

tcurrent = 0;

tt = [];
yy = [];
y0 = y00;

for i = 1:nDose
    y0(1) = y0(1) + Dose_umol_;

    [t,y] = ode15s(@(t,y) HbS_ODE(t,y,pars, Dose_mg),[0:dt:12],y0);
    y0 = y(end,:);

    tt = [tt; tcurrent+t];
    yy = [yy;y];

    tcurrent = tcurrent+12;
end
% post-dosing
[t,y] = ode15s(@(t,y) HbS_ODE(t,y,pars,0), [0:dt:addlDay*24],y0);
tt = [tt; tcurrent+t];
yy = [yy;y];
```

```
    % vars
    NrbcAll_e12cells_ = yy(:,14);
    Hcrit = 100*max(NrbcAll_e12cells_*1e12*Vrbc_L_cell_/Vblood_L_,0);
    Hb_g_dL_ = (Hcrit/100)*HbTotRBC0_uM_*64e3*1e-6*1e-1;
    dHb_g_dL_ = Hb_g_dL_ - Hb_g_dL_(1);

    AHb0rbc_umol_ = yy(:,8);
    AHb1rbc_umol_ = yy(:,9);
    AHb2rbc_umol_ = yy(:,10);
    HbCov_Pct_ = 100*(AHb1rbc_umol_ + AHb2rbc_umol_)./(eps+ AHb0rbc_umol_ +
AHb1rbc_umol_ + AHb2rbc_umol_);

    dHb_g_dL_ind(ii,:) = dHb_g_dL_';
    HbCov_Pct_ind(ii,:) = HbCov_Pct_';
    Hb_g_dL_ind(ii,:) = Hb_g_dL_';
    tsim(ii,:) = tt;
end

% plot
tt = tsim(1,:)';
HbCov = HbCov_Pct_ind;
dHb = dHb_g_dL_ind;
Hbt = Hb_g_dL_ind;

idx0 = 1;
idx4 = min(find(tt/(24*7) == 4));
idx8 = min(find(tt/(24*7) == 8));

f2 = figure('Name','Time Course','Units','Inches','Position',[1 1 10 9]);

[ts,it,is] = unique(tt/(24*7));
dHb_hi = quantile(dHb(:,it),0.95)';
dHb_lo = quantile(dHb(:,it),0.05)';
patch([ts;flipud(ts)],[dHb_hi;flipud(dHb_lo)],...
 'b','facealpha',0.3,'edgecolor','none'),hold on
plot(tt/(24*7),median(dHb,1),'color',[0.6350 0.0780 0.1840],'linewidth',2)
```

```
xlabel('Time (weeks)')
ylabel('Change in Hb (g/dL)')
set(gca,'fontsize',24)

xt_week = [0:2:Nweeks];
xticks(xt_week)
xx = axis;

plot([nDose/(7) nDose/(7)],[xx(3) xx(4)],'k--','linewidth',1),
plot([0 Nweeks],[0 0],'k--','linewidth',1),
axis([0 Nweeks xx(3) xx(4)])
xlim([0 14])

f=get(gca,'Children');
legend([f(4),f(3)],'90% Interval','Median')

saveas(f2,'duration_vp_1250mg.png')

%% plot HbCov to dHB boxplot -- Figure 4C
clear

% load vpop parameters
parset = table2array(readtable('vpop.csv','ReadVariableNames', false));
Nsample = length(parset);   % number of virtual population

% dose range and duration
DoseV = [0:250:5000];
Nweek = 8;

% initialize
HbCovAve = zeros(Nsample,length(DoseV));
HbCovMax = zeros(Nsample,length(DoseV));
HbCovMin = zeros(Nsample,length(DoseV));
DHb = zeros(Nsample,length(DoseV));
DHb4wk = zeros(Nsample,length(DoseV));

% loop over individual
```

```
parfor i = 1:Nsample
    pars = parset(i,:);

    [HbCov_Pct_Ave, HbCov_Pct_Max, HbCov_Pct_Min, DHb_g_dL_Nweek,
dHb_g_dL_4week] = run_dose_resp(pars, DoseV, Nweek);

    HbCovAve(i,:) = HbCov_Pct_Ave;
    HbCovMax(i,:) = HbCov_Pct_Max;
    HbCovMin(i,:) = HbCov_Pct_Min;
    DHb(i,:) = DHb_g_dL_Nweek;
    DHb4wk(i,:) = dHb_g_dL_4week;
end

% find coverage value yielding dHB>2 g/dL
for i = 1:Nsample
   Cov_2gdL(i) = interp1(DHb(i,:),HbCovAve(i,:),2);
end

f3 = figure('Name','Summary Dose Response', 'Units','Inches','Position',[1 1 7 9]);
boxplot(Cov_2gdL,20)
xticklabels({''})
title({'Hb Coverage for'; '2g/dL Change in Hb'})
ylabel('Hb Coverage %')
set(findobj(gca,'type','line'),'linew',2)
set(gca,'fontsize',24)

saveas(f3,'vp_dose_response.png')

%% Function for computing steady state for initial condition
function [EPO0_U_mL_, NCFUe0_e12cells_,Nret0_e12cells_, Nrbc0_e12cells_, ...
   NrbcAll0_e12cells_,RetFrac0, Hcrit0, AHb0rbc0]  = setIC_Ind(pars)
% compute steady-state value based on parameters

p = num2cell(pars);
[MolWD, ka_hr_, fbio, Vblood_L_, CLp_L_hr_, Qrbc_L_hr_, ...
Vper_L_, Qper_L_hr_, cKD1_uM_, cKD2_uM_, fup, Hcrit0, ...
HbTotRBC0_uM_, fHemolysis, kelHbp_hr_, ...
kdeathRBC0_hr_, Imaxkd, KCov_Pct_, nd, ...
```

```
vEPOmax_U_mLhr_, Krbc_e12cells_, n, kelEPO_hr_, ...
Vrbc_L_cell_, vrmin_e12cells_hr_, EmaxEPO, Kepo_U_mL_, m, ...
kCFUeRet_hr_, kret_hr_, kHbon1_uMhr_] = deal(p{:});

NrbcAll0_e12cells_ = (Hcrit0/100)*1e-12*Vblood_L_/Vrbc_L_cell_;
Nrbc0_e12cells_ = NrbcAll0_e12cells_*kret_hr_/(kret_hr_ + kdeathRBC0_hr_);
Nret0_e12cells_ = NrbcAll0_e12cells_*kdeathRBC0_hr_/(kret_hr_ + kdeathRBC0_hr_);
NCFUe0_e12cells_ = Nret0_e12cells_*kret_hr_/kCFUeRet_hr_;
y0 = [0,NCFUe0_e12cells_, Nret0_e12cells_, Nrbc0_e12cells_];

ss = fsolve(@(x) model(x,pars), y0);

EPO0_U_mL_ = ss(1);
NCFUe0_e12cells_ = ss(2);
Nret0_e12cells_ = ss(3);
Nrbc0_e12cells_ = ss(4);
NrbcAll0_e12cells_ = ss(3) + ss(4);
RetFrac0 = ss(3)/ss(4);
Hcrit0 = 100*NrbcAll0_e12cells_*1e12*Vrbc_L_cell_/Vblood_L_;
AHb0rbc0 = (kCFUeRet_hr_*NCFUe0_e12cells_*1e12*Vrbc_L_cell_*HbTotRBC0_uM_)/...
        (kdeathRBC0_hr_*(Nrbc0_e12cells_/NrbcAll0_e12cells_));

end

function y = model(x,pars)

p = num2cell(pars);
[MolWD, ka_hr_, fbio, Vblood_L_, CLp_L_hr_, Qrbc_L_hr_, ...
Vper_L_, Qper_L_hr_, cKD1_uM_, cKD2_uM_, fup, Hcrit0,  ...
HbTotRBC0_uM_, fHemolysis, kelHbp_hr_, ...
kdeathRBC0_hr_, Imaxkd, KCov_Pct_, nd, ...
vEPOmax_U_mLhr_, Krbc_e12cells_, n, kelEPO_hr_, ...
Vrbc_L_cell_, vrmin_e12cells_hr_, EmaxEPO, Kepo_U_mL_, m, ...
kCFUeRet_hr_, kret_hr_, kHbon1_uMhr_] = deal(p{:});

y = zeros(4,1);
% vEPO_U_mLhr_ = vEPOmax_U_mLhr_*Krbc_e12cells_^n/(Krbc_e12cells_^n +
NrbcAll_e12cells_^n);
```

```
% dEPO_U_mL_ = vEPO_U_mLhr_ - kelEPO_hr_*EPO_U_mL_;
y(1) = vEPOmax_U_mLhr_*Krbc_e12cells_^n/(Krbc_e12cells_^n + (x(3)+x(4))^n) -
kelEPO_hr_*x(1);

% dNCFUe_e12cells_ = kbirthCFUe_e12cells_hr_ - kCFUeRet_hr_*NCFUe_e12cells_;
% kbirthCFUe_e12cells_hr_ = vrmin_e12cells_hr_*(1 + ...
%                    EmaxEPO*EPO_U_mL_^m/(Kepo_U_mL_^m + EPO_U_mL_^m));
y(2) = vrmin_e12cells_hr_*(1 + EmaxEPO*x(1)^m/(Kepo_U_mL_^m + x(1)^m)) -
kCFUeRet_hr_*x(2);

% dNret_e12cells_ = kCFUeRet_hr_*NCFUe_e12cells_ - kret_hr_*Nret_e12cells_;
y(3) = kCFUeRet_hr_*x(2) - kret_hr_*x(3);

% HbCov=0 at baseline (no drug)
% kdeathRBC_hr_ = kdeathRBC0_hr_*(1 - Imaxkd*HbCov_Pct_^nd/(KCov_Pct_^nd +
HbCov_Pct_^nd));
% dNrbc_e12cells_ = kret_hr_*Nret_e12cells_ - kdeathRBC_hr_*Nrbc_e12cells_;
y(4) = kret_hr_*x(3) - kdeathRBC0_hr_*x(4);
end

%% Function for running Dose Response

function [HbCov_Pct_Ave, HbCov_Pct_Max, HbCov_Pct_Min, DHb_g_dL_Nweek,
dHb_g_dL_4week] = run_dose_resp(pars, DoseV, Nweek)
% generate dose response
% HbCov_Pct_Ave, Max Min -- Hb Covered
% DHb_g_dL_Store -- Change in Hb compared to baseline at Nweek
% dHb_g_dL_4week -- Change in Hb compared to baseline at 4 week

%% params
p = num2cell(pars);
[MolWD, ka_hr_, fbio, Vblood_L_, CLp_L_hr_, Qrbc_L_hr_, ...
Vper_L_, Qper_L_hr_, cKD1_uM_, cKD2_uM_, fup, Hcrit0, ...
HbTotRBC0_uM_, fHemolysis, kelHbp_hr_, ...
kdeathRBC0_hr_, Imaxkd, KCov_Pct_, nd, ...
vEPOmax_U_mLhr_, Krbc_e12cells_, n, kelEPO_hr_, ...
Vrbc_L_cell_, vrmin_e12cells_hr_, EmaxEPO, Kepo_U_mL_, m, ...
kCFUeRet_hr_, kret_hr_, kHbon1_uMhr_] = deal(p{:});
```

```
 %% set IC
[EPO0_U_mL_, NCFUe0_e12cells_,Nret0_e12cells_, Nrbc0_e12cells_, ...
   NrbcAll0_e12cells_,RetFrac0, Hcrit0,AHb0rbc0]  = setIC_Ind(pars);

y0 = zeros(14,1);
y0(8) = AHb0rbc0; y0(11) = EPO0_U_mL_; y0(12) = NCFUe0_e12cells_;
y0(13) = Nret0_e12cells_;  y0(14) = Nrbc0_e12cells_;
y00 = y0;

%% Run Dose Resp
nDose = Nweek*7*2;     %BID dosing
dt = 2;              % hour

HbCov_Pct_Ave = zeros(1,length(DoseV));
HbCov_Pct_Max = zeros(1,length(DoseV));
HbCov_Pct_Min = zeros(1,length(DoseV));

DHb_g_dL_Nweek = zeros(1,length(DoseV));
dHb_g_dL_4week = zeros(1,length(DoseV));

for i = 1: length(DoseV)

Dose_mg = DoseV(i);
Dose_umol_ = Dose_mg*1e3/MolWD;

tcurrent = 0;

tt = [];
yy = [];
y0 = y00;

for j = 1:nDose
    y0(1) = y0(1) + Dose_umol_;

    [t,y] = ode15s(@(t,y) HbS_ODE(t,y,pars,Dose_mg),[0:dt:12],y0);
    y0 = y(end,:);

    tt = [tt; tcurrent+t];
```

```
    yy = [yy;y];

    tcurrent = tcurrent+12;
end

%% Hb (g/dL)
NrbcAll_e12cells_ = yy(:,14);
Hcrit = 100*max(NrbcAll_e12cells_*1e12*Vrbc_L_cell_/Vblood_L_,0);
Hb_g_dL_ = (Hcrit/100)*HbTotRBC0_uM_*64e3*1e-6*1e-1;
dHb_g_dL_ = Hb_g_dL_ - Hb_g_dL_(1);

AHb0rbc_umol_ = yy(:,8);
AHb1rbc_umol_ = yy(:,9);
AHb2rbc_umol_ = yy(:,10);
HbCov_Pct_ = 100*(AHb1rbc_umol_ + AHb2rbc_umol_)./(eps+ AHb0rbc_umol_ +
AHb1rbc_umol_ + AHb2rbc_umol_);

% average over last 24 hours
lenday = length([0:dt:24]);
HbCov_Pct_Ave(i) = mean(HbCov_Pct_(end-lenday:end));
HbCov_Pct_Max(i) = max(HbCov_Pct_(end-lenday:end));
HbCov_Pct_Min(i) = min(HbCov_Pct_(end-lenday:end));

DHb_g_dL_Nweek(i) = mean(Hb_g_dL_(end-lenday:end)) - Hb_g_dL_(1);

idx4 = find(tt== 4*7*24);
dHb_g_dL_4week(i) = mean(Hb_g_dL_(idx4-lenday:idx4))- Hb_g_dL_(1);
end
end
```